PHYSICS

1) A particle is projected from a horizontal plane (x-z plane) such that its velocity vector at time t is given by $\vec{V} = a\hat{i} + (b - ct)\hat{j}$. Its range on the horizontal plane is given by :-

(1) $\frac{ba}{c}$

(2) $\frac{2ba}{c}$

(3) $\frac{3ba}{c}$

(4) None

2) Two balls are thrown with the same initial speed at angles 30° and 60° with the horizontal. What will be the ratio of maximum heights attained by them :-

(1) 1 : 2

(2) 1 : 1

(3) 2 : 1

(4) 1 : 3

3) It is possible to project a particle on a level ground with a given velocity in 2 possible ways so as to have the same range R. The product of corresponding times of flight is then :

(1) $\frac{R}{g}$

(2) $\frac{2R}{g}$

(3) $\frac{R}{2g}$

(4) None of these

4) A projectile is projected with speed u at an angle θ with vertical find out speed when velocity vector becomes perpendicular to initial velocity.

(1) u cot θ

(2) u cos θ

(3) u sin θ

(4) u tan θ

5)

It was calculated that a shell when fired from a gun with a certain velocity and at an angle of

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elevation $\frac{5\pi}{36}$ rad should strike a given target in the same horizontal plane. In actual practice, it was found that a hill just prevented the trajectory. At what angle of elevation should the gun be fired to hit the target.

(1) $\frac{5\pi}{36}$rad

(2) $\frac{11\pi}{36}$rad

(3) $\frac{7\pi}{36}$rad

(4) $\frac{13\pi}{36}$rad

6)

A projectile is fired from the surface of the earth with a velocity of 5 m/s and angle θ with the horizontal. Another projectile fired from another planet with a velocity of 3 m/s at the same angle follows a trajectory which is identical with the trajectory of the projectile fired from the earth. The value of the acceleration due to gravity on the planet is (in m/s^2) is: (given g = 9.8 m/s^2)

(1) 3.5

(2) 5.9

(3) 16.3

(4) 110.8

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7) A particle is projected from the ground with an initial speed of v at an angle θ with horizontal. The average velocity of the particle between its point of projection and highest point of trajectory is :-

(1) $\frac{v}{2}\sqrt{1+2\cos^2\theta}$

(2) $\frac{v}{2}\sqrt{1+\cos^2\theta}$

(3) $\frac{v}{2}\sqrt{1+3\cos^2\theta}$

(4) v cos θ

8) A particle moves in x-y plane according to the law x = kt, y = kt (1 – αt) where k and α are positive constant and t is time. The trajectory of the particle is :

(1) y = kx

(2) $y = x - \frac{\alpha x^2}{k}$

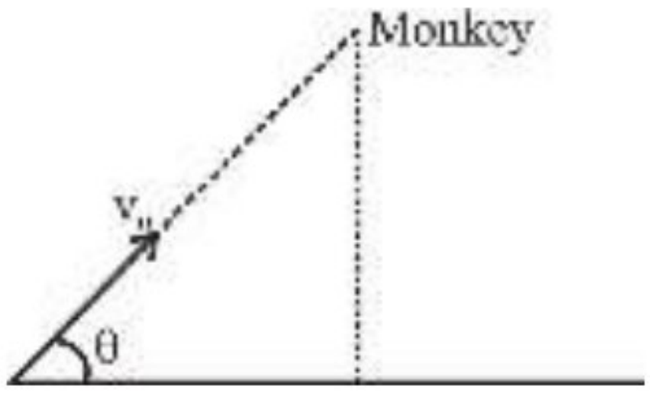
(3) $y = -\frac{\alpha x^2}{k}$

(4) $y = \alpha x$

9) A bullet is fired at an angle θ with horizontal aiming towards a monkey. The bullet will hit the

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monkey only if :- 

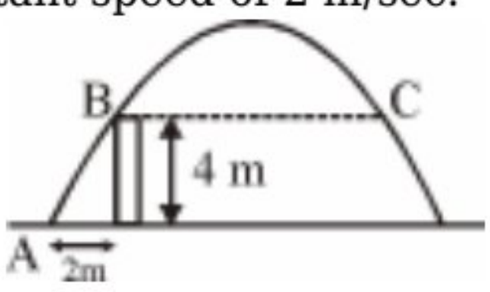
(1) Monkey fall down at the same instant the gun is fired.

(2) Monkey fall down after the gun is fired

(3) Monkey fall down before the gun is fired

(4) Monkey remains stationary

10) A particle is projected from point A. A fly is standing at a tower at point B as shown in figure. When particle passes from B, fly starts flying in horizontal direction with constant speed of 2 m/sec.



Both will meet at the same horizontal level at point C. Then : ($g = 10\ m/sec^2$):-

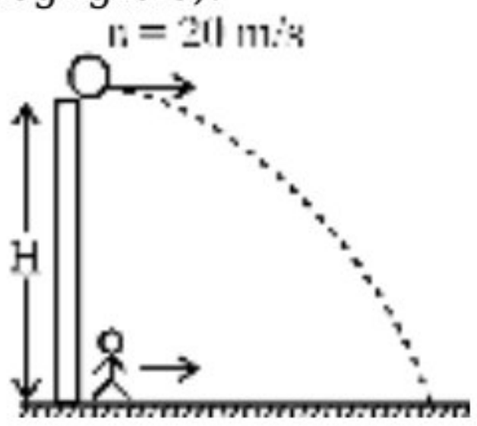
(1) Angle of projection is $\tan^{-1}\left(\frac{9}{2}\right)$

(2) Angle of projection is $\tan^{-1}(2)$

(3) Angle of projection is 45º

(4) None of these

11) A ball is thrown horizontally from the top of a tower of height H with velocity 20 m/s. A man is standing at the foot of tower starts running at the same instant when ball is thrown. What should be the velocity of man to catch the ball before striking to the ground. (Consider height of man negligible):-



(1) 10 m/s

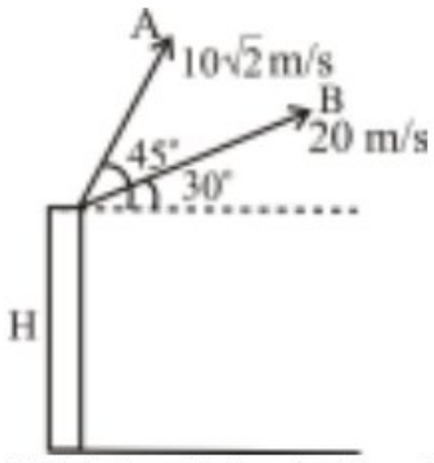
(2) $\sqrt{2gH}$

(3) 20 m/s

(4) none of these

12) Two balls are thrown from a tower as shown in the figure :-

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Which of the ball will hit the ground first :-

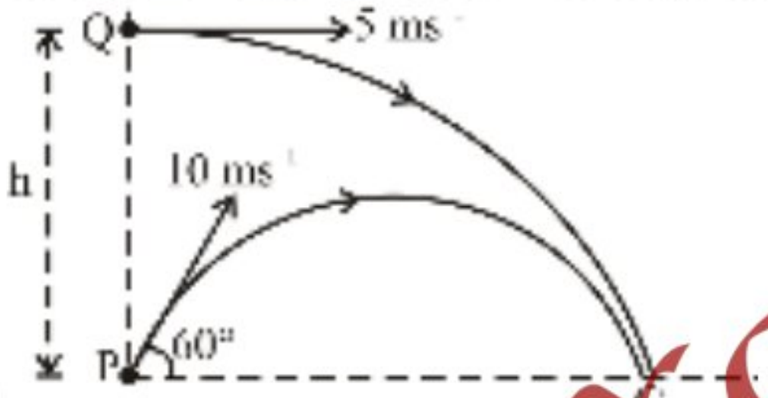
(1) A

(2) B

(3) Both will strike at the same time

(4) Depends on height of tower

13) A particle P is projected from ground with an initial speed of 10m/s at an angle of 60° with the horizontal. From what height h should an another particle Q be projected horizontally with velocity 5 m/s so that both the particles collide at ground at point C, if both are projected simultaneously? (g = 10 m/s^2)



(1) 10 m

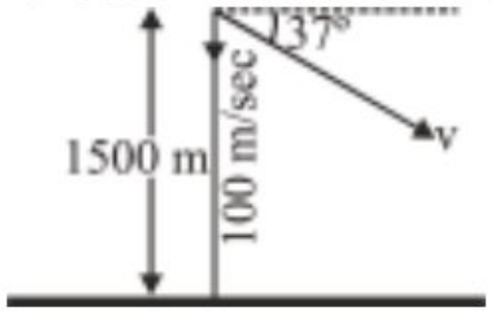
(2) 30 m

(3) 15 m

(4) 25 m

14) A particle is projected from top of a tower of height 1500 m and at a velocity v making an angle 37º with the horizontal and its vertical downward component is 100m/s as shown in figure. The distance from the foot of the tower where it strikes the ground will be : -

(g = 10 m/sec^2)



(1) 4000/3 m

(2) 5000/3 m

(3) 2000 m

(4) 3000 m

15) Two particles are simultaneously projected in opposite directions horizontally from a given point in space where gravity 'g' is uniform. If u_1 and u_2 be their initial speeds then the time 't' after which their velocities are mutually perpendicular is given by

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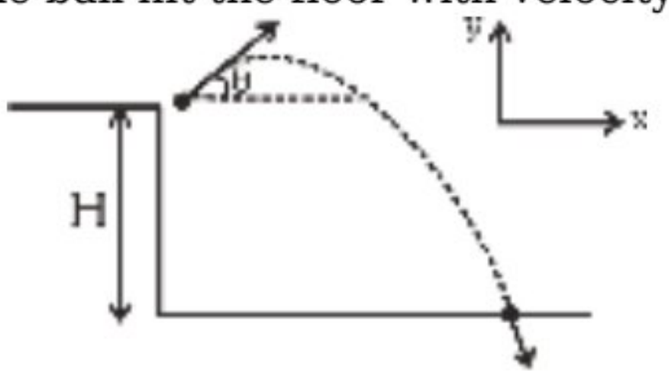
(1) $\frac{\sqrt{u_1 u_2}}{g}$

(2) $\frac{\sqrt{u_1^2 + u_2^2}}{g}$

(3) $\frac{\sqrt{u_1(u_1 + u_2)}}{g}$

(4) $\frac{\sqrt{u_2(u_1 + u_2)}}{g}$

16) A ball was thrown from height H and the ball hit the floor with velocity $10(\hat{i} - \hat{j})$ after 1.5 sec of its projection. The angle of projection θ is :



(1) $\theta = \tan^{-1}\left(\frac{3}{2}\right)$

(2) $\theta = \tan^{-1}\left(\frac{1}{2}\right)$

(3) $\theta = \tan^{-1}(2)$

(4) $\theta = \tan^{-1}(1)$

17) A bus begins to move with an acceleration of 1 ms^{-2} and at the same time, a man who is 48 m behind the bus starts running at 10 ms^{-1} to catch the bus. The man will be able to catch the bus after :-

(1) 6 s

(2) 5 s

(3) 3 s

(4) 8 s

18) A train is moving towards east and a car is along north, both with same speed. The observed direction of car to the passenger in the train is :-

(1) East-north direction

(2) West-north direction

(3) South-east direction

(4) None of these

19) Two cars are moving in the same direction with a speed of 30 km h^{-1}. They are separated from each other by 5 km.Third car moving in the opposite direction meets the two cars after an interval

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of 4 minutes. What is the speed of the third car?

(1) 30 km h^{-1}

(2) 35 km h^{-1}

(3) 40 km h^{-1}

(4) 45 km h^{-1}

20) A person walks up a stalled escalator in 90 sec. He is carried in 60s, when standing on the same escalator which is now moving. The time he would take to walk up the moving escalator will be :–

(1) 27 s

(2) 72 s

(3) 18 s

(4) 36 s

21) A ship A is moving Westwards with a speed of 10 km/h and a ship B 100 km South of A, is moving Northwards with a speed of 10 km/h. The time after which the distance between them becomes shortest, is :-

(1) 5 h

(2) $5\sqrt{2}$h

(3) $10\sqrt{2}$h

(4) 0 h

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22) Two persons A and B running on a track in the same direction observe a car. A says that the car is moving in east direction and B says that the car is moving in north direction. They contradict the direction but say that magnitude is same. If the speed of B is double that of the speed of A, then the true direction of the car will be :-

(1) $\theta = \tan^{-1}\left(\frac{1}{2}\right)$ North of East

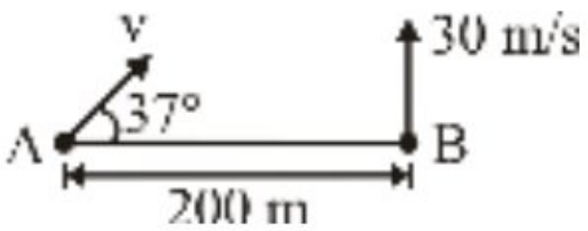
(2) $\theta = \tan^{-1}\left(\frac{1}{2}\right)$ South of East

(3) $\theta = \tan^{-1}\left(\frac{1}{3}\right)$ North of East

(4) None of these

23)

Two particles A and B are moving in horizontal plane as shown in figure at t = 0, then time after which A will catch to B will be-



(1) 4 s

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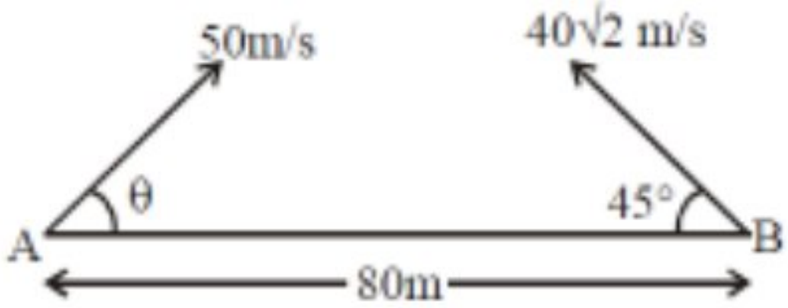
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(2) 5 s

(3) 2 s

(4) 8 s

24) Two particle are projected from position 'A' and 'B' separated by a distance of 80 m on horizontal ground. If both the particle collides in mid air. Select the correct statement [Both particles are projected simultaneously]



(1) They will collide if angle of projection for 'A' is 37°

(2) Particle will collide between the two points 'A' and 'B'.

(3) Particle will collide after 1 sec of projection

(4) They will collide if angle of projection for 'A' is 53°

25)

A man is walking at a speed 3 m/s rain drops are falling vertically with a speed 3 m/s

(i) What is the velocity of rain drop with respect to the man?

(ii) At what angle from vertical, the man should hold his umbrella?

(1) 2.42 m/s, 30° in forward direction

(2) 4.24 m/s, 45° in forward direction

(3) 1.24 m/s, 60° in forward direction

(4) None of these

26) Rain is falling vertically downward with a speed of 35 m/s. Wind starts blowing after some time with a speed of 12 m/s in East to West direction. The direction in which a boy standing at the place should hold his umbrella is:

(1) $\tan^{-1}\left(\frac{12}{37}\right)$ w.r.t. wind

(2) $\tan^{-1}\left(\frac{12}{35}\right)$ w.r.t. rain

(3) $\tan^{-1}\left(\frac{12}{35}\right)$ w.r.t. wind

(4) $\tan^{-1}\left(\frac{12}{37}\right)$ w.r.t. rain

27) A river 2 km wide is flows at the rate of 2km/h. A boatman who can row a boat at a speed of 4 km/h in still water, goes a distance of 2 km upstream and then comes back. The time taken by him to complete his journey is

(1) 60 min

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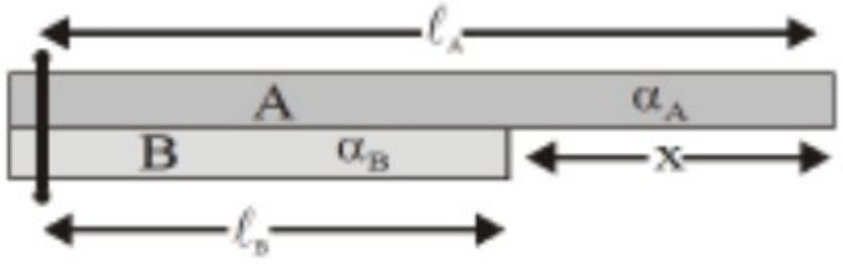
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(2) 70 min

(3) 80 min

(4) 90 min

28) Bars of two different metals are bolted together, as shown in figure. The distance x does not change with temperature if



(1) $\frac{\ell_A}{\ell_B} = \frac{\alpha_A}{\alpha_B}$

(2) $\frac{\ell_A}{\ell_B} = \frac{\alpha_B}{\alpha_A}$

(3) $\frac{\ell_A^2}{\ell_B^2} = \frac{\alpha_A}{\alpha_B}$

(4) $\frac{\ell_A^2}{\ell_B^2} = \frac{\alpha_B}{\alpha_A}$

29) Two thermometers 'X' & 'Y' shows boiling point & freezing point of water as 220°X & 20°X and 120°Y & –40°Y respectively. If 'X' shows 100°X, then find the reading in 'Y' thermometer.

(1) 25°Y

(2) 50°Y

(3) 20°Y

(4) 24°Y

30) The apparent coefficient of expansion of a liquid when heated in a copper vessel is C and when heated in a silver vessel is S. If A is the coefficient of linear expansion of copper, the coefficient of linear expansion of silver is :-

(1) $\frac{C+S-3A}{3}$

(2) $\frac{C+3A-S}{3}$

(3) $\frac{S+3A-C}{3}$

(4) $\frac{C+S+3A}{3}$

31) Calculate the time required to heat 10 kg of water from 10°C to 60°C using an immersion heater rated 5000 W. Assume that 60% of the power input is used to heat the water. Specific heat capacity of water = 4200 J/kg-k

(1) 3.2 min

(2) 11.6 min

(3) 20.3 min

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(4) 50 min

32) A steel ball of mass 0.1 kg falls freely from a height of 10 m and bounces to a height of 5.4 m from the ground. If the dissipated energy in this process is absorbed by the ball, the rise in its temperature is :
(specific heat of steel=460 J/kg/°C,g=10 m/s^2)

(1) 0.01°C

(2) 0.1°C

(3) 1°C

(4) 1.1°C

33) Steam at 100°C is added slowly to 1400 gm of water at 16°C until the temperature of water is raised to 80°C. The mass of steam required :-

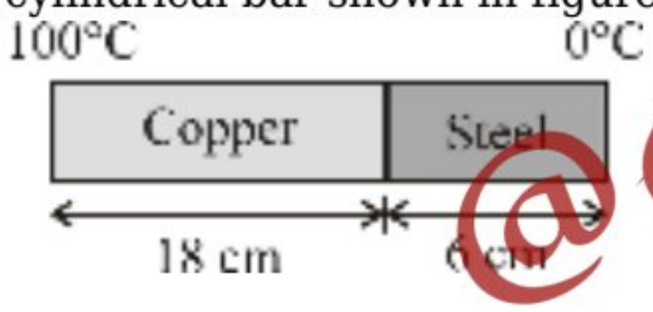
(1) 120 gm

(2) 240 gm

(3) 160 gm

(4) 320 gm

34) The coefficient of thermal conductivity of copper is nine times that of steel. In the composite cylindrical bar shown in figure, what will be the temperature at the junction of copper and steel ?



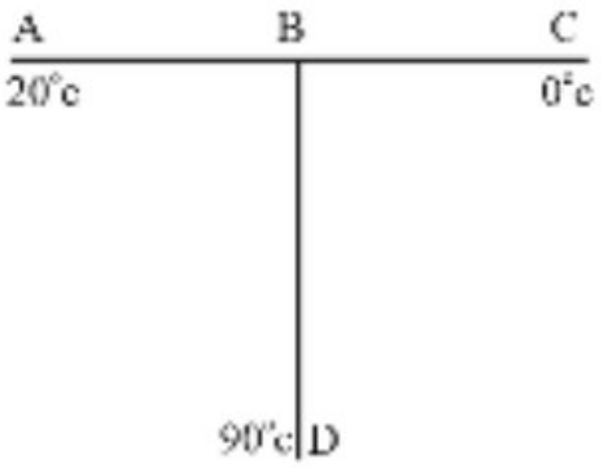
(1) 75°C

(2) 67°C

(3) 33°C

(4) 25°C

35) Three conducting rods of same material and cross-section are shown in figure. Temperature of A, D and C are maintained at 20°C, 90°C and 0°C. Find the ratio of length BD and BC if there is no heat flow in AB.



(1) $\frac{2}{7}$

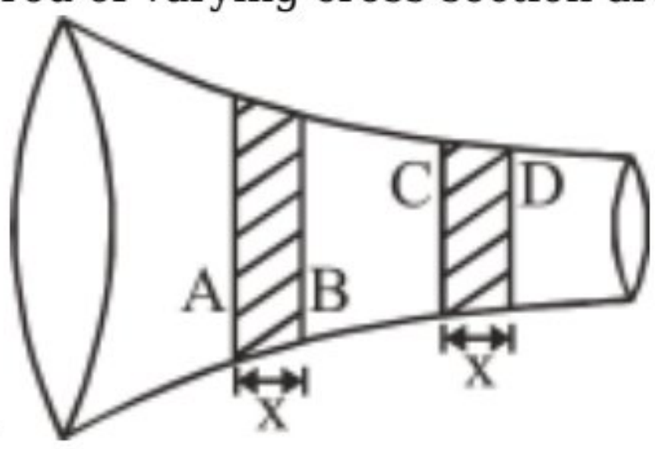
(2) $\frac{1}{2}$

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(3) $\frac{7}{2}$

(4) $\frac{1}{3}$

36) Two ends of a conducting rod of varying cross-section are maintained at 200°C & 0°C



respectively. In steady state :-

(1) Temp difference across AB and CD are equal

(2) Temp difference across AB is greater than that of across CD

(3) Temp difference across AB is less than that of across CD

(4) Temp difference may be equal or different depending in the thermal conductivity of rod.

37) A cup of tea cools from 80°C to 60°C in one minute. The ambient temperature is 30°C. In cooling from 60°C to 50°C, it will take :

(1) 50 sec.

(2) 90 sec.

(3) 60 sec.

(4) 48 sec.

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38) For a black body at temperature 727°C, its radiating power is 60 watt and temperature of surrounding is 227°C. If temperature of black body is changed to 1227°C then its radiating power will be :-

(1) 304 W

(2) 320 W

(3) 240 W

(4) 120 W

39) The rectangular surface of area 8cm × 4 cm of a black body at a temperature of 127°C emits energy at the rate of E. If the length and breadth of the surface are each reduced to half of the initial value and the temperature is raised to 327°C, the rate of emission of energy will become.

(1) $\frac{3}{8}E$

(2) $\frac{81}{16}E$

(3) $\frac{9}{16}E$

(4) $\frac{81}{64}E$

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40) If the temperature of stars of colours Red, Green and Blue are respectively T_R, T_G and T_B then

(1) $T_B > T_R > T_G$

(2) $T_B < T_G < T_R$

(3) $T_B > T_G > T_R$

(4) $T_G > T_B > T_R$

41) The power radiated by a black body is P and it radiates maximum energy around the wavelength λ_0. If the temperature of the black body is now changed so that it radiates maximum energy around wavelength $\frac{\lambda_0}{4}$, the power radiated by it well be :-

(1) 4P/3

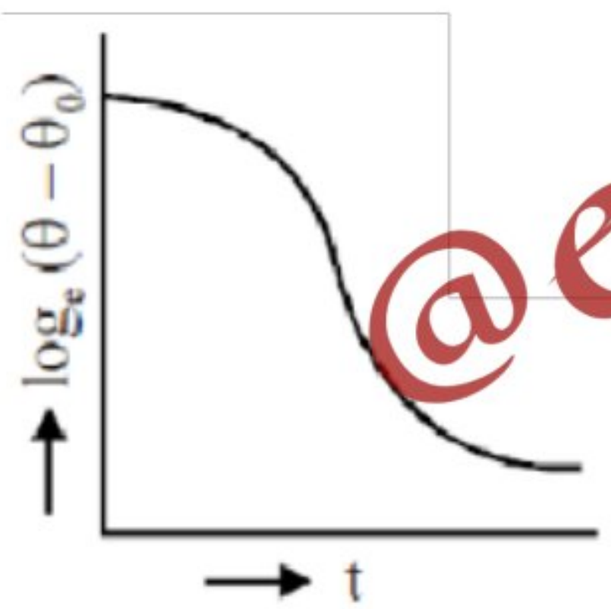
(2) 256 P

(3) P/16

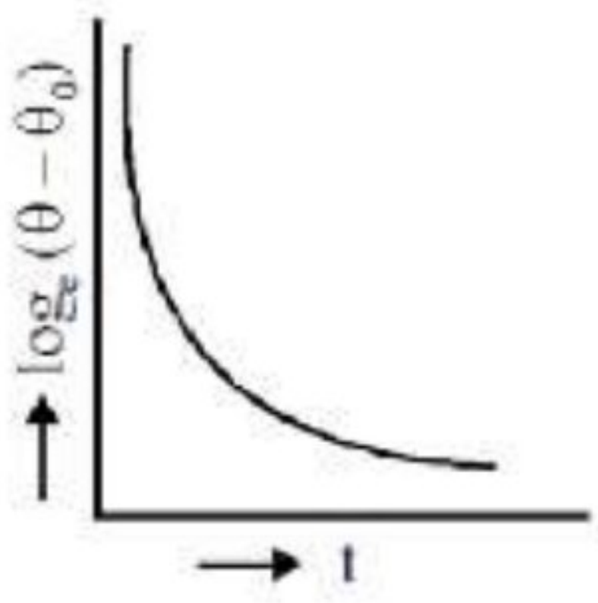
(4) 256/81 P

42) A liquid in a beaker has temperature θ at time t and θ_0 is temperature of surroundings, then according to Newton's law of cooling, correct graph between $\log_e(\theta - \theta_0)$ and t is:

(1)



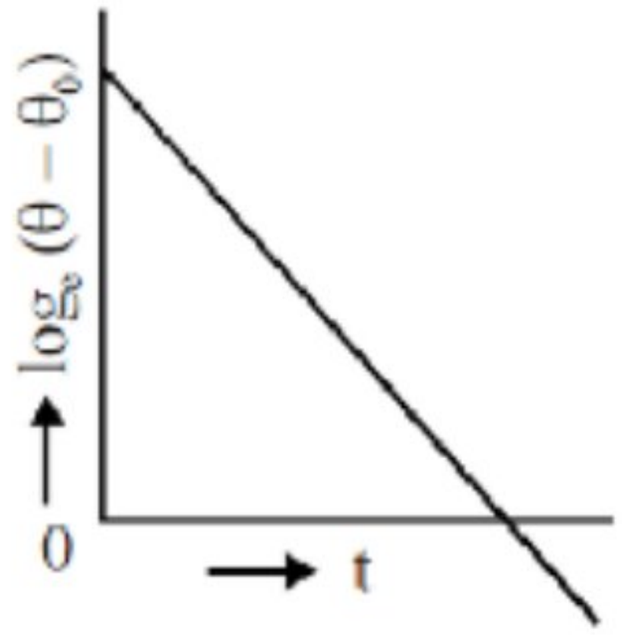
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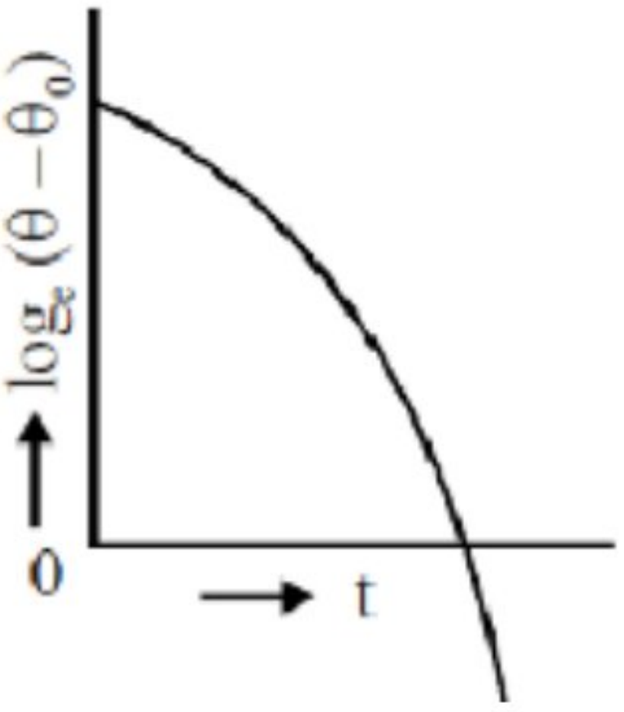


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(3) 

(4) 

43) A rigid tank contains 35 kg of nitrogen at 6 atm. Sufficient quantity of oxygen is supplied to increase pressure to 9 atm while temp remain constant. Amount of oxygen supplied to the tank is :-

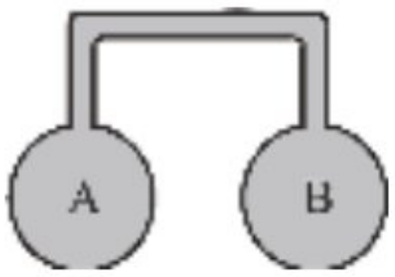
(1) 5 kg

(2) 10 kg

(3) 20 kg

(4) 40 kg

44) Two spherical vessel of equal volume, are connected by a narrow tube. The apparatus contains an ideal gas at one atmosphere and 300K. Now if one vessel is immersed in a bath of constant temperature 600K and the other in a bath of constant temperature 300K. Then the common pressure will be :-



(1) 1

(2) $\frac{4}{5}$atm

(3) $\frac{4}{3}$atm

(4) $\frac{3}{4}$atm

45) A gas contained in a box of 0.1 m^3 at atmospheric pressure is connected to another vessel of 0.09

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m^3. Consequent change in pressure is X mm of Hg. Then X in metre is -

(1) 0.4

(2) 0.5

(3) 0.36

(4) 0.3

CHEMISTRY

1) Unpaired electron is present in :-

(1) NO_2^+ and CaO_2

(2) KO_2 and Na_2O_2

(3) KO_2 only

(4) CaO_2 only

2) Among the following electron deficient compound is :-

(1) CCl_4

(2) PCl_5

(3) OF_2

(4) BCl_3

3) Hypervalent compounds is (are) :

(i) PCl_5 (ii) SF_6

(iii) BF_3 (iv) ClF_3

Correct option is :

(1) i, ii, iii

(2) ii, iii, iv

(3) i, ii, iv

(4) All of these

4) The shape of a molecule which has EB_5L_2 moleculer formula :- {E = Central atom, B = Terminal atom, L = Loan pair}

(1) Square pyramidal

(2) Pentagonal planar

(3) Octahedral

(4) Pentagonal bipyramidal

5) Strongest bond is formed by head on overlapping of :-

(1) 2s and 2p orbital

(2) 2p and 2p orbital

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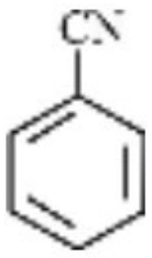
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(3) 2s and 2s orbital

(4) All of them have equal strength

6)


has

(1) 7σ + 4π bonds

(2) 7σ + 3π bonds

(3) 12σ + 6π bonds

(4) 13σ + 5π bonds

7) In which of the following molecules no. of lone pairs and bond pairs on central atoms are not equal :-

(1) H_2O

(2) IF_5

(3) OF_2

(4) SCl_2

8) In which of the following reaction, the hybridisation of the central atom changes from sp^2 to sp^3 ?

(1) $NH_3 + H^+ \rightarrow NH_4^+$

(2) $SiF_4 + 2F^- \rightarrow SiF_6^{-2}$

(3) $BF_3 + F^- \rightarrow BF_4^-$

(4) $H_2O + H^+ \rightarrow H_3O^+$

9) Find the incorrect statement :-

(1) Number of hybrid orbitals produced is equal to the number of atomic orbitals that get hybridised.

(2) Hybrid orbitals are equivalent in energy and shape

(3) Hybrid orbitals are more effective in forming sigma bonds than pure atomic orbitals

(4) In hybridisation only half filled atomic orbitals participates

10)

Correct order of C – O bond length in -

(1) $CO_3^{-2} < CO_2 < CO$

(2) $CO_2 < CO_3^{-2} < CO$

(3) $CO < CO_3^{-2} < CO_2$

(4) $CO < CO_2 < CO_3^{-2}$

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11) The geometry of ammonia molecule can be best described as :-

(1) Nitrogen at one vertex of a regular tetrahedron, the other three vertices being occupied by three hydrogens

(2) Nitrogen at the centre of the tetrahedron, three of the vertices being occupied by three hydrogens

(3) Nitrogen at the centre of an equilateral triangle, three corners being occupied by three hydrogens

(4) Nitrogen at the junction of a T, three open ends being occupied by three hydrogens

12) The correct statement regarding SO_2 molecule is

(1) Two pπ-dπ bonds

(2) Molecule has two lone pair, two σ-bonds and two π-bonds

(3) Two pπ-pπ bonds

(4) One pπ-pπ and one pπ-dπ bond

13) Which of the following species represent maximum no. of pπ – dπ bond ?

(1) SO_3

(2) SO_2

(3) ClO_4^-

(4) NO_3^-

14) The shape of covalent molecule MX_3 is :-

(1) triangular

(2) T-shape

(3) pyramidal

(4) either of these depending on number of lone pairs of electron on M

15) Number and type of bonds between two carbon atoms in CaC_2 are :-

(1) one sigma (σ) and one pi(π) bonds.

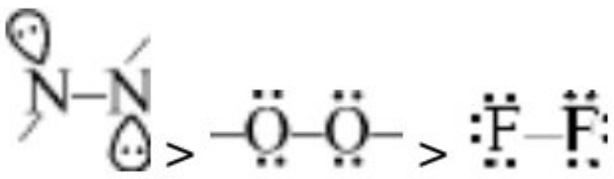
(2) one σ and two π bonds.

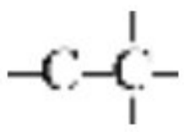
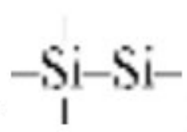
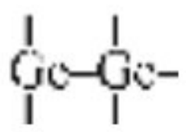
(3) one σ and one and a half π bonds.

(4) one σ bond.

16) Which of the following order of BE is not correct ?

(1) Cl–Cl > Br–Br > F–F

(2)  > >

(3)  >  > 

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(4) O=O < H–H < F–F

17) Maximum bond angle is present in :-

(1) H_2O

(2) H_3O^+

(3) NH_4^+

(4) CO_2

18) Which of the following molecule does not contains bond angle of 90° –

(1) IF_7

(2) $SiCl_4$

(3) XeF_4

(4) PCl_5

19) The paramagnetism of O_2 is due to presence of unpaired electrons in the MO :-

(1) σ*2s

(2) π*2px

(3) π*2py

(4) Both 2 & 3

20) Which of the following diamagnetic species having only π bond in vapour phase :

(1) O_2 and C_2

(2) O_2^{-2}

(3) B_2 and O_2

(4) C_2 and B_2

21) **Assertion :** Bonding molecular orbital possesses lower energy then either of the atomic orbitals.
Reason : Electrons placed in a bonding molecular orbital tend to hold the nuclei together and stabilise a molecule.
Both are correct & reason explains assertion.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

22) Which one of the following properties is **not** shown by NO ?

(1) It combines with oxygen to form nitrogen dioxide

(2) It's bond order is 2.5

(3) It is diamagnetic in gaseous state

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(4) It is a neutral oxide

23) The correct order of the strength of H bonds is

(1) $HF > H_2O > NH_3$
(2) $NH_3 > H_2O > HF$
(3) $H_2O > NH_3 > HF$
(4) $HF > NH_3 > H_2O$

24) o-nitrophenol is more volatile than para-nitrophenol due to -

(1) intramolecular H-bonding in o-nitrophenol and intermolecular H-bonding in p-nitrophenol
(2) intermolecular H-bonding in o-nitrophenol and intramolecular H-bonding in p-nitrophenol
(3) more stronger intramolecular H-bonding in o-nitrophenol as compared to p-nitrophenol
(4) more stronger intermolecular H-bonding in o-nitrophenol as compared to p-nitrophenol

25) Which of the following statement is not correct

(1) The crystal lattice of ice mostly formed by covalent as well as hydrogen bonds.
(2) O-nitrophenol can be easily steam distilled whereas p-nitrophenol cannot be
(3) Vander Walls forces are responsible for the formation of molecular crystals.
(4) Boiling point of NH_3 is greater than SbH_3

26) Identify the incorrect order :-

(1) HF > HCl > HBr > HI ; Acidic nature
(2) Li^+ (aq) < Be^{+2}(aq.) ; hydrated radius
(3) LiF > NaF > KF > RbF ; Lattice energy
(4) Cl > F > Br > I ; e^- affinity

27) Molecule showing correct order of solubility is :-
(i) $CaSO_4 < MgSO_4$ (ii) $KHCO_3 > NaHCO_3$
(iii) $NaNO_3 < KNO_3$ (iv) LiF > LiI
(v) $CH_3OH < CH_3OCH_3$

(1) (i), (ii) and (iii)
(2) (iii), (iv) and (v)
(3) (i) and (ii)
(4) (i) only

28) Which of following order of lattice energy is incorrect ?

(1) $Al_2O_3 > Na_2O$
(2) $MgF_2 < CaF_2$
(3) LiF > NaF
(4) LiF > LiI

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29) In which of the following pairs, first compound has higher lattice energy, Hydration energy and solubility ?

(1) $MgCl_2$, $BeCl_2$

(2) $CaCO_3$, $MgCO_3$

(3) BeF_2, CaF_2

(4) $Be(OH)_2$, $Mg(OH)_2$

30) Which of the following pair of compounds have covalent as well as ionic bond :-

(1) KCN & K_2O

(2) CaC_2 & $CaCl_2$

(3) RbO_2 & NaOH

(4) HCN & H_2O

31) Select the not incorrect for dipole moment order ?

(1) $NH_3 > NCl_3$

(2) $XeF_2 = XeF_4 < SO_2$

(3) $P(CH_3)_2 (CF_3)_3 < P (CF_3)_2(CH_3)_3$

(4) 1 & 2 both are not incorrect

32) What is correct order of dipole moment (μ) ?

(1) Benzene < Borazine

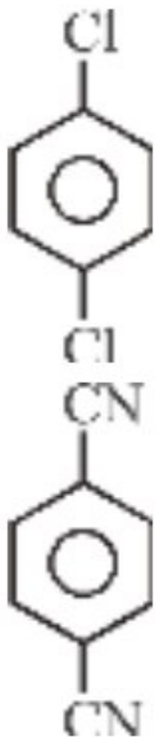
(2) $SiF_4 > SF_4$

(3) $SO_2 > CO_2$

(4) $NF_3 > NH_3$

33)

Which molecule is polar ?

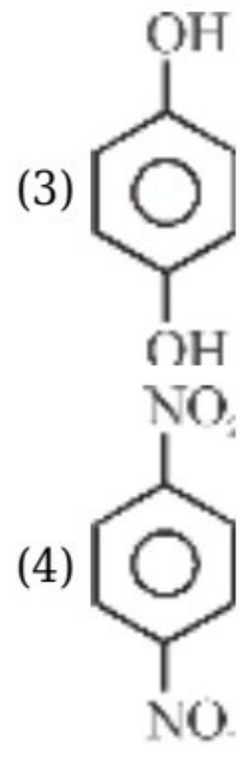
(1) 

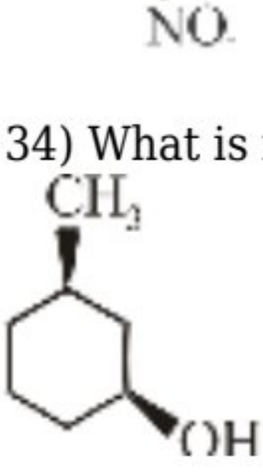
(2)

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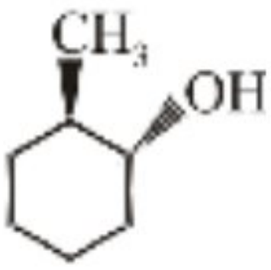
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(3) 

(4) 

34) What is relation between the following pairs

 &  are :-

(1) Identical

(2) Geometrical isomer

(3) Positional isomer

(4) (2) & (3) both

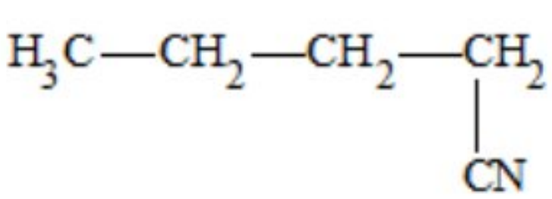
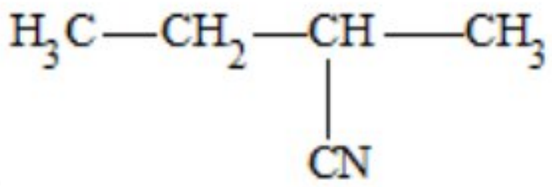
35) Minimum number of carbon atom in alkane to show position isomerism is :-

(1) 3

(2) 4

(3) 5

(4) 6

36)  &  isomerism is

(1) Position isomerism

(2) Chain isomerism

(3) Functional isomerism

(4) Both 1 & 2

37) $CH_3CHOHCH_2CHO$ and $CH_3CH_2CH_2COOH$ constitute a pair of :-

(1) Position isomers

(2) Metamers

(3) Optical isomers

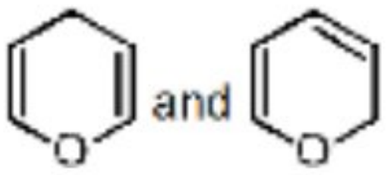
(4) Functional isomers group

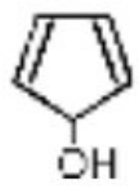
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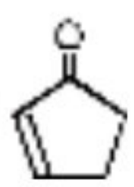
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38) Which of the following pairs of structures do not represent isomers -

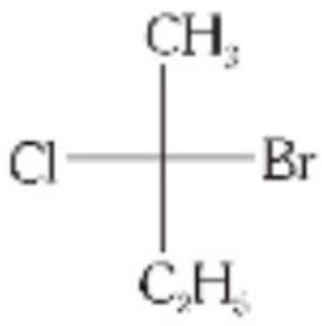
(1)  and

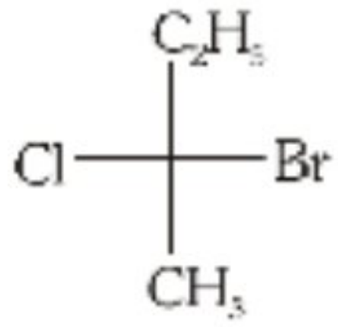
(2)  and 

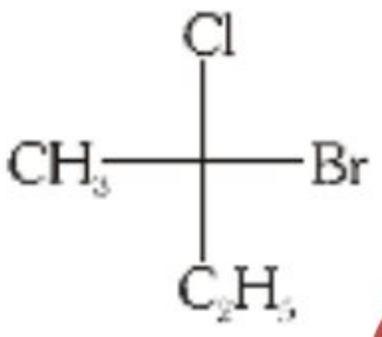
(3)  and 

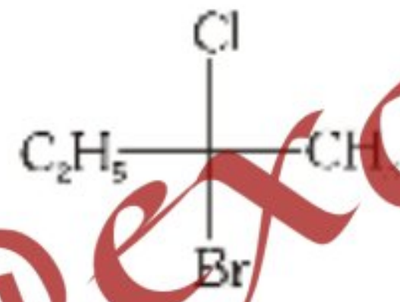
(4)  and 

39) Consider the following structures (A), (B), (C) and (D) -

(A) 

(B) 

(C) 

(D) 

Which of the following statements is not correct

(1) B and C are identical

(2) A and B are enantiomers

(3) A and C are enantiomers

(4) B and D are enantiomers

40) $CH_3-NH-C_2H_5$, $CH_3-CH_2-CH_2-NH_2$ and $(CH_3)_3N$ show which type of isomerism :

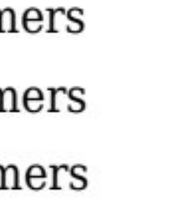
(1) Position

(2) Functional

(3) Chain

(4) None

41) How many structural isomers are possible for following compound when one H-atom is replaced



by one chlorine atom :-

(1) 2

(2) 4

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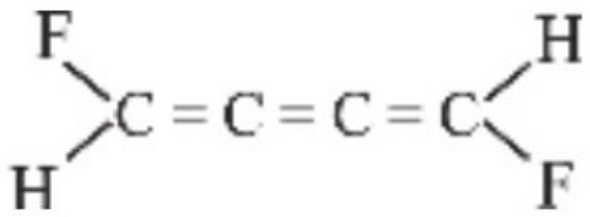
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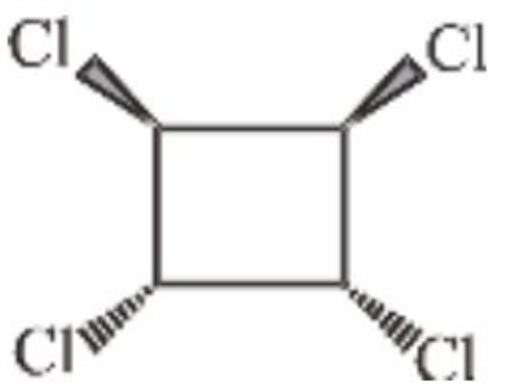
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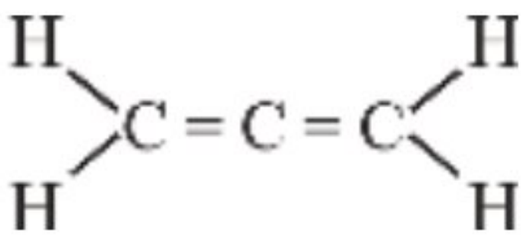
(3) 6

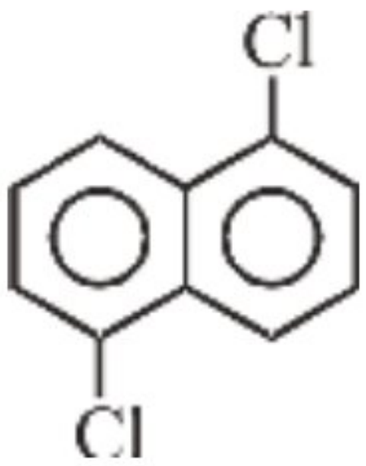
(4) 8

42) Which of the following has plane of symmetry (POS) but not centre of symmetry (COS) ?

(1) 

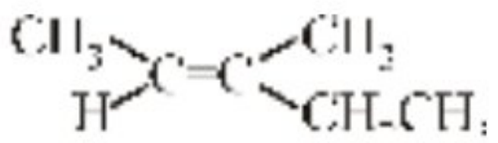
(2) 

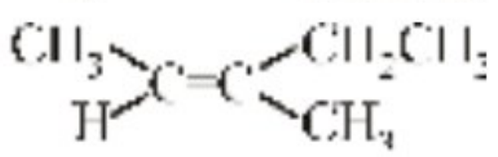
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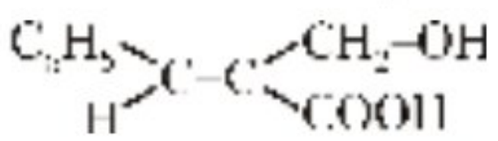
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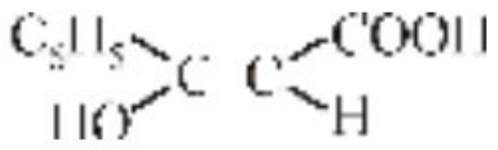
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43) The Z-isomer among the following is :-

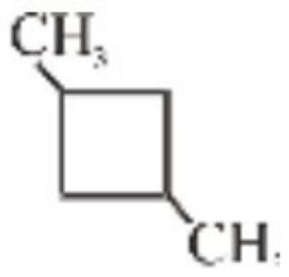
(1) 

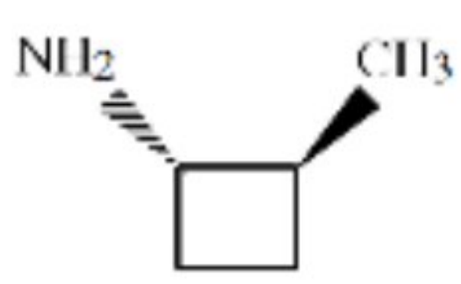
(2) 

(3) 

(4) 

44) Which of the following is optically active ?

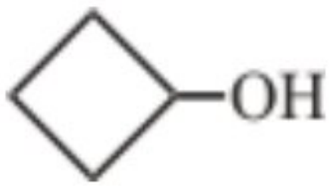
(1) 

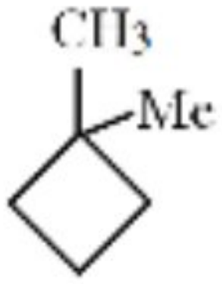
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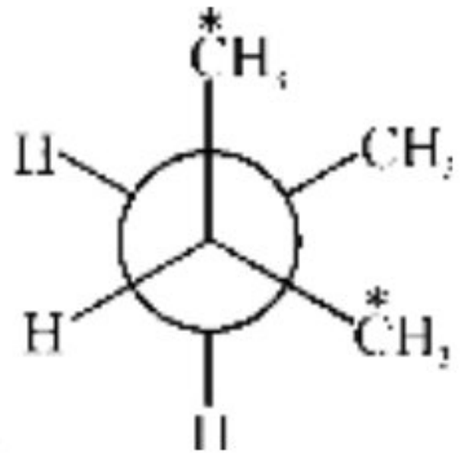
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NOVA

(3) 

(4) 



45) What is bond angle between star marked (*) carbon atoms

(1) 120°

(2) 60°

(3) 109°28'

(4) Can not determine

BIOLOGY

1) Read the following point carefully.
(a) Nitrogen fixation ability
(b) Oxygenic photosynthesis
(c) Colonies are generally surrounded by gelatinous sheath
(d) Cause water bloom
These points are related with-

(1) Eubacteria

(2) Euglenoids

(3) Green algae

(4) Cyanobacteria

2) Select incorrect statement ?

(1) Heterotrophic bacteria are most abundant in nature.

(2) The cyanobacteria are unicellular, colonial or filamentous

(3) Halophiles are found in extreme salty areas

(4) Mycoplasma can not survive without oxygen.

3) In Whittaker's 'Five Kingdom Classification', eukaryotes were assigned to :-

(1) Only two of the five kingdoms

(2) Only three of the five kingdoms

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(3) Only four of the five kingdoms

(4) All the five kingdoms

4) Based on the sequences of 16s rRNA genes, Woese found that _____ kingdoms, naturally cluster in to _____ domains.

(1) 5, 3

(2) 3, 5

(3) 6, 3

(4) 3, 6

5) What is the criteria used by Whittaker for classification?
(i) Cell structure
(ii) Body organization
(iii) Mode of nutrition
(iv) Reproduction
(v) Phylogenetic relationship
(vi) Biochemical difference
(vii) Physiological character

(1) All except (v) and (vi)

(2) All except (v) and (vii)

(3) All except (vi) and (vii)

(4) All except (iii) and (iv)

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6) Select the total number of disease from the following caused by bacteria :
Mumps, Smallpox, Citrus canker, Cholera, Typhoid, Tetanus, Sleeping sickness, Malaria

(1) 2

(2) 4

(3) 5

(4) 6

7) **Statement I:-** Heterotrophic bacteria are most abundant in nature. The majority are important decomposers.
Statement II:- The mycoplasma are organisms that completely lack a cell wall. They are the smallest non-living cells known and can survive without oxygen

(1) Both statement I and statement II are false

(2) Statement I is correct but statement II is false

(3) Statement I is incorrect but statement II is true

(4) Both statement I and statement II are true

8) In prokaryotes, the genetic material is

(1) linear DNA with histones

(2) circular DNA with histones

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(3) linear DNA without histones
(4) circular DNA without histones

9) Choose the correct match

(1) *Gonyaulax* – Red sea
(2) *Euglena* – Chlorophyll a & c
(3) Desmids – Chrysophytes
(4) *Gymnodinium* – Hemicellulosic plates in wall

10) Which of the following are acellular

(1) Viruses
(2) Viroids
(3) Prions
(4) All of these

11) Which of the following is not a feature of Bentham and Hookers classification ?

(1) It was based on actual observations
(2) It was a natural form of classification
(3) It was based on floral features
(4) It was a phylogenetic classification

12) In modern classification of organisms according to the 5 kingdom system now the *Chlorella, Chlamydomonas, Paramoecium, Amoeba* are put in the kingdom :-

(1) Monera
(2) Protista
(3) Plantae
(4) Mycota

13) Viroids differ from viruses in having;

(1) DNA molecules without protein coat
(2) RNA molecules with protein coat
(3) RNA molecules without protein coat
(4) DNA molecules with protein coat

14) Which of the following structure protects bacteria from phagocytosis ?

(1) Cell wall
(2) Glycocalyx
(3) Flagella
(4) Pili

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15) Which of the following do not have peptidoglycan ?

(1) Halophiles
(2) Methanogen
(3) Thermoacidophiles
(4) All of these

16) The walls are embedded with silica and thus the walls are indestructible in :-

(1) Slime moulds
(2) Euglenoids
(3) Diatoms
(4) Dinoflagellates

17)

Select the option which includes all the correct ones only :-
(A) Diatoms → Chief producer of ocean
(B) Dinoflagellate → Without flagella
(C) Slime mould → Decomposer nature
(D) Euglenoids → Sometime behave like predator

(E) Protozoa → Unicellular prokaryotes
(1) A, B and C
(2) A, B, C and D
(3) B, C, D and E
(4) A, C and D

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18) A parasitic protozoan that has flagella is

(1) *Gonyaulax*
(2) A malaria causing organism
(3) *Amoeba*
(4) *Trypanosoma*

19) Given below are two statements :
Statement I : Protista forms a link with the others dealing with plants, animals and fungi.
Statement II : Under suitable conditions slime moulds form an aggregation called plasmodium.
Choose the correct answer from the options given below.

(1) Both statement I and II are correct
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct
(4) Both statement I and II are incorrect

20)

Stored food material of fungi :-

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(1) Cellulose
(2) Starch
(3) Glycogen and starch
(4) Glycogen and oil

21) Septate and branched mycelium is found in :-

(1) Basidiomycetes
(2) Deuteromycetes
(3) Ascomycetes
(4) All the above

22) **Assertion :-** Fungi prefer to grow in cold and humid areas.
Reason :- Low temperature favours the metabolic activities.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

23)

Fungi show asexual reproduction by all of the following kinds of spores except

(1) conidia
(2) oospores
(3) sporangiospores
(4) zoospores

24) Select the correctly matched pair :-

(1) Oomycetes : *Puccinia*
(2) Basidiomycetes : *Alternaria*
(3) Phycomycetes : *Rhizopus*
(4) Deuteromycetes : *Penicillium*

25) Which of the following is odd ?

(1) Toad stool
(2) *Puccinia*
(3) *Alternaria*
(4) Mushroom

26) Which of the following are not included in the five kingdom system of classification ?
(a) Viruses (b) Viroids (c) Lichens

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(1) Only a & b

(2) Only b

(3) Only b & c

(4) All a, b & c

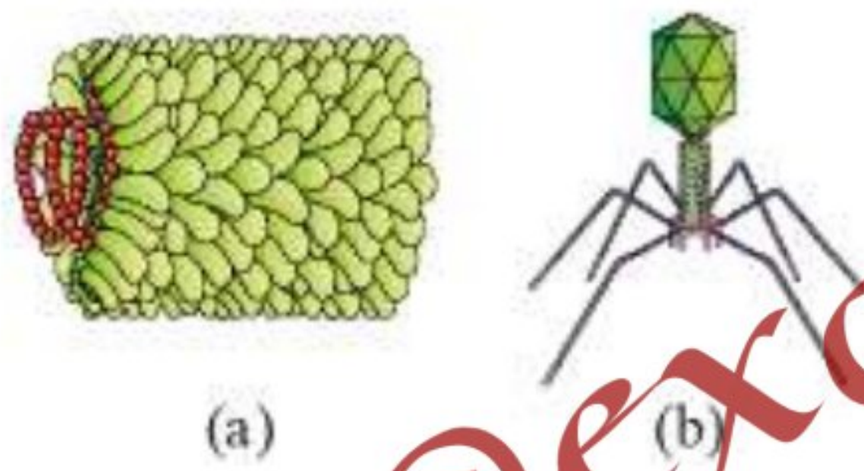
27) Read the following statements :-
(A) Lichens do not grow in polluted areas.
(B) Viroids have low molecular weight DNA
(C) Virus are inert outside their specific host cell.
(D) Mumps, small pox and influenza disease caused by Bacteria.
Which statements are **correct** ?

(1) A and B

(2) A and C

(3) A, B and C

(4) A, B and D



28)
Host cell for the viruses (a) and (b) could be:-

(1) Prokaryotic cell and Eukaryotic cell

(2) Plant cell and Eukaryotic cell

(3) Eukaryotic cell and prokaryotic cell

(4) Plant cell and animal cell

29) Which of the following are heterotrophic organism?

(1) Ascomycetes

(2) Diatoms

(3) Algae

(4) Gymnosperms

30) Which is incorrect:

(1) T.O. Diener discovered virus in 1981

(2) RNA of viroid was of low molecular weight

(3) Lichens are symbiotic association

(4) Lichens are pollution indicator.

31) All of these are obligate parasites, except

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(1) Viroid

(2) Tobacco mosaic virus

(3) *Rhizopus*

(4) T4 phage

32) Find the incorrect match w.r.t possession of nucleic acid material :

(1)	Plant viruses	ssRNA usually
(2)	Animal viruses	dsDNA/dsRNA/ssRNA
(3)	Plant viruses	dsDNA usually
(4)	Bacteriophages	dsDNA usually

(1) 1

(2) 2

(3) 3

(4) 4

33) Match the column I with column II :-

Particles/Micro organisms		**Diseases**	
(i)	Viroid	(a)	CJD
(ii)	Virus	(b)	Citrus canker
(iii)	Bacteria	(c)	Potato spindle tuber
(iv)	Prion	(d)	Mumps

(1) i - a, ii - b, iii - d, iv - c

(2) i - c, ii - d, iii - a, iv - b

(3) i - c, ii - d, iii - b, iv - a

(4) i - a, ii - c, iii - b, iv - d

34) The boundaries of kingdom 'x' is not well-defined. Which of the following organism belongs to kingdom 'x' :-

(1) Amoeba

(2) Anabaena

(3) Yeast

(4) Halophiles

35) **Assertion:** Bacteria reproduce mainly by binary fission.
Reason: Under unfavourable condition they produce endospores.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

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(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

36) **Assertion (A) :** Majority of bacteria are autotrophs.
Reason (R) : Chemoheterotrophic nutrition is absent in bacteria.

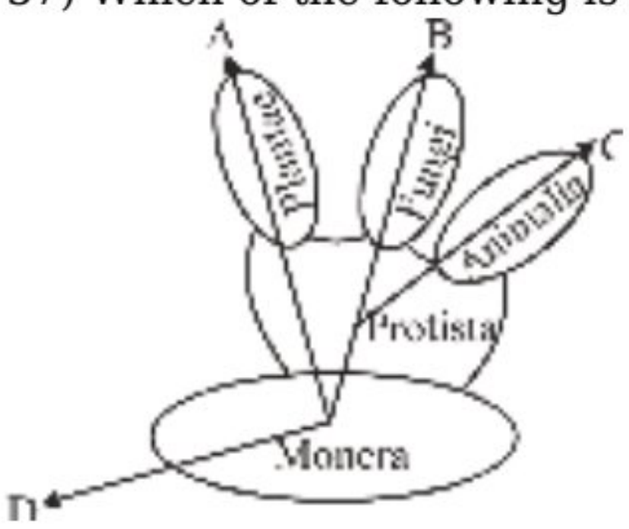
(1) Both Assertion & Reason are False.

(2) Assertion is True but the Reason is False.

(3) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(4) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

37) Which of the following is not correct option for given representing line B in given below diagram.



(1) Absorptive

(2) Heterotrophy

(3) Decomposer

(4) Autotrophic

38) Identify A, B and C in given diagram.



(1) A = *Mucor*, B = *Aspergillus*, C = *Agaricus*

(2) A = *Mucor*, B = *Agaricus*, C = *Aspergillus*

(3) A = *Agaricus*, B = *Mucor*, C = *Aspergillus*

(4) A = *Agaricus*, B = *Aspergillus*, C = *Mucor*

39) Given below are two statements :
Statement-I : Eubacteria are characterised by the presence of a rigid cell wall and oxygenic photosynthesis.
Statement-II : The Mysoplasma are organism that completely lack a cell membrane.
In the light of the above statements, choose the correct answer from the option given below.

(1) Both statement I and II are false

(2) Statement I is correct but statement II is false

(3) Both statement I and II are true

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(4) Statement I is false and II is true

40) Match the following columns and select the correct option–

	Column I		**Column II**
a.	*Rhizopus*	(i)	Ascospore
b.	*Neurospora*	(ii)	Basidiospore
c.	Bracket fungi	(iii)	Zygospore

(1) a–(i), b–(ii), c–(iii)

(2) a–(iii), b–(i), c–(ii)

(3) a–(iii), b–(ii), c–(i)

(4) a–(i), b–(iii), c–(ii)

41) Which group of organism have soap box like structure, indestructible wall, float passively and microscopic nature ?

(1) Dinoflagellates

(2) Euglenoids

(3) Chrysophytes

(4) Protozoans

42) Which of following is not the function of mesosome ?

(1) Cell respiration

(2) Cell wall formation

(3) DNA replication

(4) Flagella formation

43) Who was the first to attempt a more scientific basis of classification ?

(1) Linnaeus

(2) Aristotle

(3) Whittaker

(4) Bentham and Hooker

44)

Read the following terms carefully.
Cilia, Flagella, Zygote, Diatomite, Photosynthesis, Pellicle, Peptidoglycan wall, Bioluminescence, Multicellular, heterocyst, decomposer, Holozoic.

How many terms are not related with kingdom Protista:-

(1) 4

(2) 5

(3) 3

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(4) 6

45) Match Column-I and Column-II.

Column - I		Column - II	
(a)	Slime mould	(i)	Organism are mostly marine and photosynthetic and produced toxins.
(b)	Chrysophytes	(ii)	Consist of dual mode of nutrition (mixotrophic)
(c)	Dinoflagellates	(iii)	Filtration of oils and syrups
(d)	Euglenoids	(iv)	The spores are dispersed by air currents

Choose the **correct** answer from option and given below.

	(a)	(b)	(c)	(d)
(1)	i	ii	iv	iii
(2)	iv	iii	i	ii
(3)	iii	iv	ii	i
(4)	ii	iv	iii	i

(1) 1

(2) 2

(3) 3

(4) 4

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46) **Statement I :** In competitive inhibition, inhibitor closely resembles the substrate in its molecular structure.
Statement II : Malonate structure resemble with succinate.

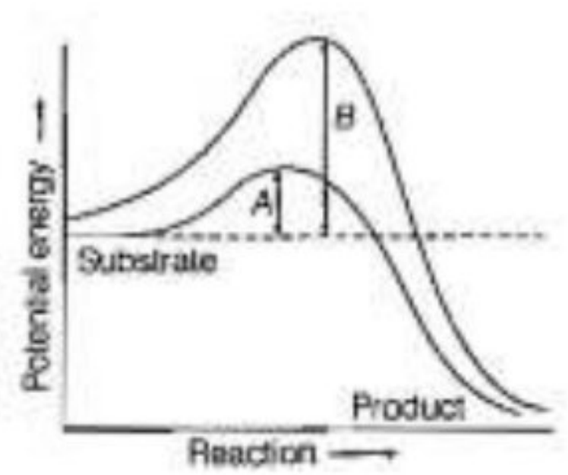
(1) Both statements are correct

(2) Statement I is correct & II is incorrect

(3) Statement I is incorrect & II is correct

(4) Both statements are incorrect

47) Which of the following describes the graph correctly?



(1) Endothermic reaction with energy A in the presence of enzyme and B in the absence of enzyme

(2) Exothermic reaction with energy A in the presence of enzyme and B in the absence of enzyme

(3) Endothermic reaction with reaction A in the absence of enzyme and B in the presence of enzyme

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(4) Exothermic reaction with energy A in the absence of enzyme and B in the presence of enzyme

48) Which one of the following statements is correct, with reference to enzymes ?

(1) Holoenzyme = Apoenzyme + Coenzyme
(2) Holoenzyme = Apoenzyme + Cofactor
(3) Apoenzyme = Holoenzyme + Cofactor
(4) 1 & 2 both

49) Match the following columns.

	Column - I (Enzymes)		**Column - II (Characteristics)**
A.	Dehydrogenases	1.	Interconversion of optical, geometrical positional isomers
B.	Ligases	2.	Group transfer
C.	Isomerases	3.	Oxidoreduction between two substrates
D.	Hydrolases	4.	Linking together of two bonds
E.	Transferases	5.	Hydrolysis of bonds

	A	**B**	**C**	**D**	**E**
(1)	5	4	1	2	3
(2)	4	3	5	2	1
(3)	5	4	2	3	1
(4)	3	4	1	5	2

(1) 1
(2) 2
(3) 3
(4) 4

50) **Assertion (A):** Enzymes through their active sites, catalyse reactions at high rate.
Reason (R): An active site of an enzyme is a pocket formed always in the quaternary structure of protein into which substrate fits.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

51) Which factor is responsible for inhibition of enzymatic process during feedback?

(1) Substrate
(2) Enzyme

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(3) End product

(4) Temperature

52) Which of the following factors can influence enzymatic activity ?

(1) Temperature

(2) pH

(3) Substrate concentration

(4) All of the above

53)

Which of the following statements is **not** correct ?

(1) Almost all proteins are enzymes

(2) All enzymes are not proteins

(3) Enzymes have tertiary structure of protein

(4) Ribozymes are made up of nucleic acids

54) The concentration of substrate at which rate of reaction of an enzyme attains half of its maximum velocity. It is represent as :-

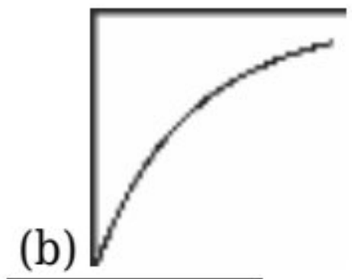
(1) Turn over number

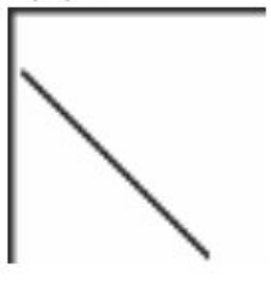
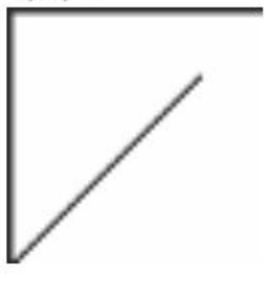
(2) Activation energy

(3) K_m (Michelis Menten Constant)

(4) V_{max}

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55) (a)  (b) 

(c)  (d) 

Out of four curves which of the following shows of substrate concentration on progression of enzyme catalysed reaction ?

(1) (d)

(2) (b)

(3) (c)

(4) (a)

56)

$CO_2 + H_2O \rightarrow H_2CO_3$ (carbonic acid)

Which one is incorrect about the above reaction?

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(1) Without enzyme, the rate of H_2CO_3 formation is 200 molecules per hour.

(2) When carbonic anhydrase catalyses the same reaction, there is no change in the rate of H_2CO_3 formation.

(3) The reaction catalysed by the enzyme shows speed with about 600,000 molecules being formed/second (10 million times more rate)

(4) The enzyme which catalyse the reaction occurs in cytoplasm of certain cells

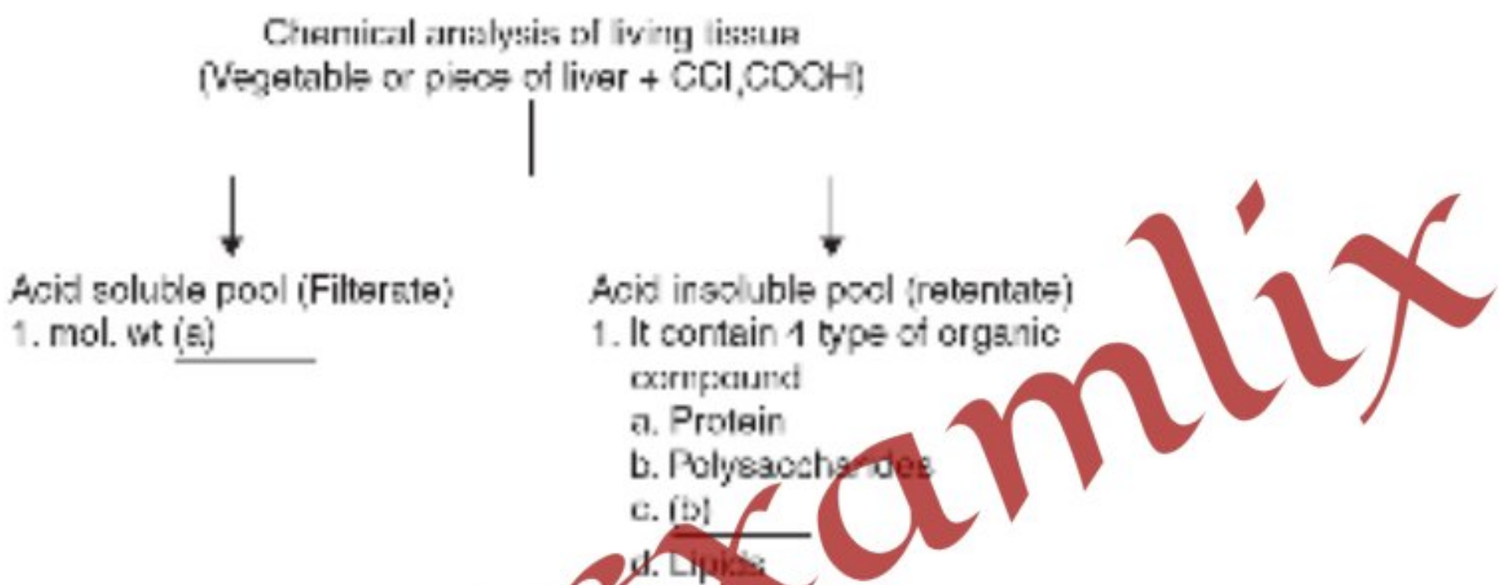
57) Sulpha drugs are used for the control of bacterial pathogens, because they cause :-

(1) Competitive inhibition of folic acid synthesis

(2) Allosteric inhibition of folic acid synthesis

(3) Feedback inhibition of folic acid synthesis

(4) Irreversible inhibition of folic acid synthesis



58)

Fill in the blanks in (1) and (2) in the above fl owchart.

(1) a → greater than 1000 Dalton; b → Nucleotide

(2) a → 18–800 Da ; b → Nucleic acid

(3) a → greater than 10,000 Da ; b → Oligosaccharides

(4) a → greater than 1000 Dalton ; b → Nucleoside

59) Match the columns I and II, and choose the correct combination from the options given :

Column-I (Component)	**Column-II (% of the total cellular mass)**
(a) Ions	(i) 1
(b) Lipids	(ii) 2
(c) Carbohydrates	(iii) 3
(d) Nucleic acids	(iv) 5-7
(e) Proteins	(v) 10-15

(1) a-i, b-ii, c-iii, d-iv, e-v

(2) a-ii, b-iii, c-i, d-v, e-iv

(3) a-iii, b-i, c-ii, d-iv, e-v

(4) a-iv, b-ii, c-iii, d-v, e-i

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60) Which of the following is a toxin and a secondary metabolite derived from the list:
Alkaloids, Carotenoids, Curcumin, Abrin, and Morphine

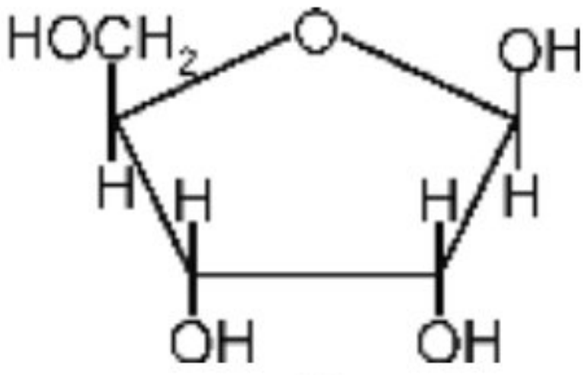
(1) Morphine

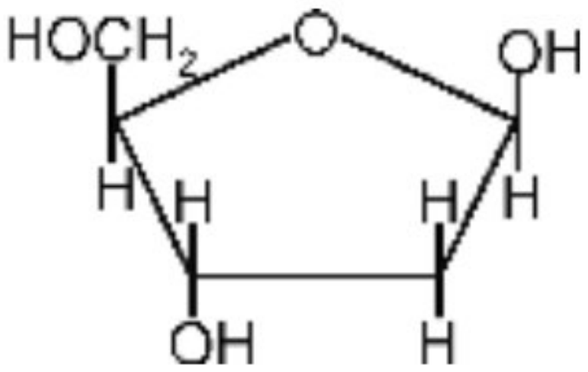
(2) Abrin

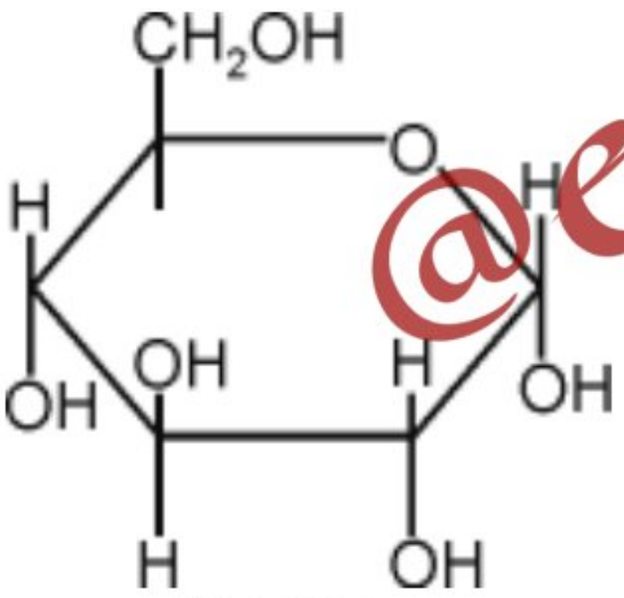
(3) Carotenoids

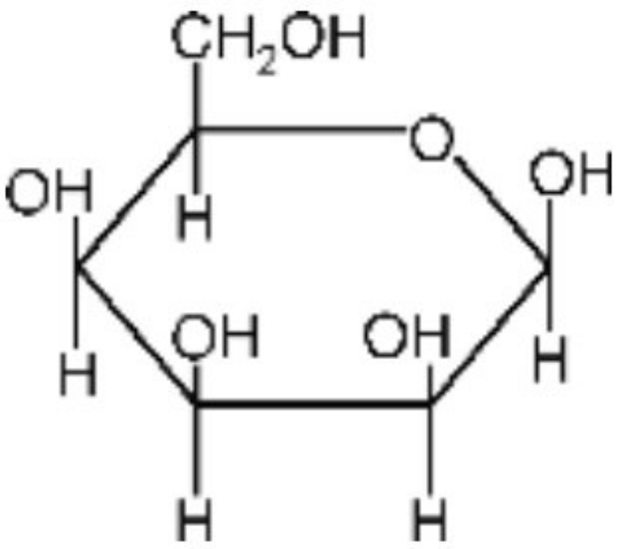
(4) Curcumin

61) Which structure is correct for the de-oxyribose sugar?

(1) 

(2) 

(3) 

(4) 

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62) Glycosidic bond at place of branching in starch and glycogen has _____.

(1) $\beta - 1,4$ linkage

(2) $\alpha - 1,6$ linkage

(3) $\alpha - 1,2$ linkage

(4) $\alpha - 1,4$ linkage

63) Which of the following statements is/are incorrect?

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I. Left end of a polysaccharide is called non-reducing end, while right end is called reducing end. II. Starch and glycogen are branched molecules. III. Starch and glycogen are the reserve food materials of plants and animals, respectively. IV. Starch can hold iodine molecules in its helical secondary structure, but cellulose being non-helical, cannot hold iodine.

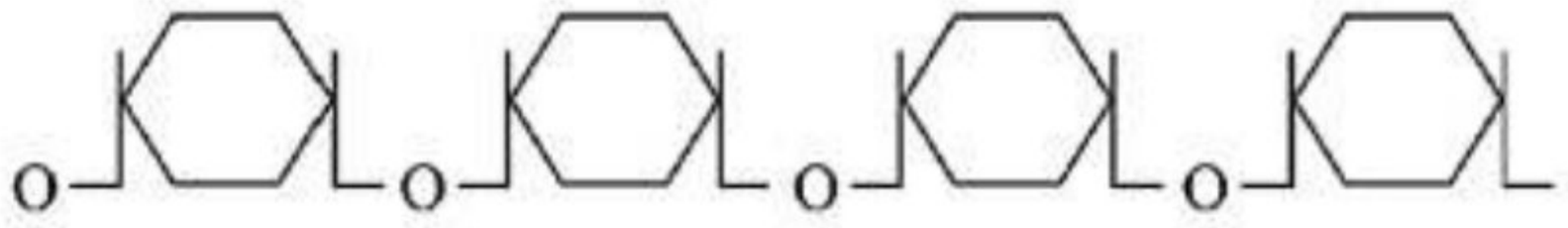
(1) I and II

(2) All statements are incorrect

(3) Only IV

(4) None of the above

64) Identify A and B bonds in the following diagrammatic representation of a portion of glycogen.



Choose the correct option.

(1) A = $1,6\alpha$-glycosidic bonds,
B = $1,4\alpha$-glycosidic bonds

(2) A = $1,1\alpha$-glycosidic bonds,
B = $1,1\alpha$-glycosidic bonds

(3) A = $1,4\alpha$-glycosidic bonds,
B = $1,4\alpha$-glycosidic bonds

(4) A = $1,4\alpha$-glycosidic bonds,
B = $1,6\alpha$-glycosidic bonds

65) Assertion (A): All monosaccharides are reducing sugar.
Reason (R): Monosaccharides contains free aldehyde or free ketonic group.

(1) Both assertion and reason are true and reason is the correct explanation of the assertion

(2) Both assertion and reason are true and reason is not the correct explanation of the assertion

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(3) Assertion is true but the reason is false

(4) Both assertion and reason are false

66) The number of reducing ends in a glycogen molecule would be

(1) Equal to the number of branches plus one

(2) Equal to the number of branch points

(3) One

(4) Two, one on the left side and another on the right side

67) Which sugar occurs only in mammals ?

(1) Trehalose

(2) Galactose

(3) Lactose

(4) Mannose

68) Select the statement which holds true for lipids

(1) Arachidonic acid has 20C excluding carboxyl carbon

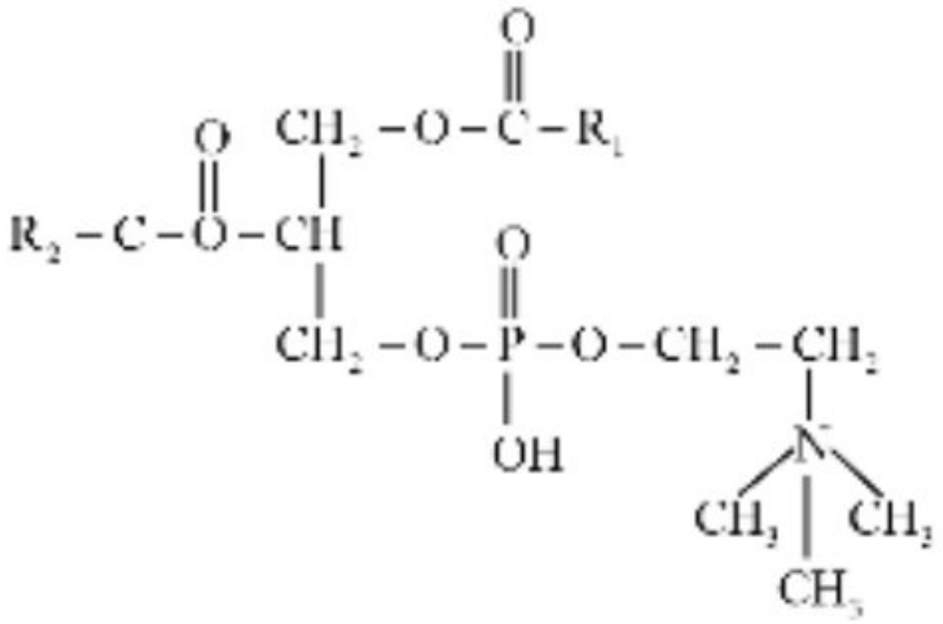
(2) Palmitic acid has 18C carbon excluding carboxyl carbon

(3) Glycerol is trihydroxy propane

(4) Neural tissues have lipids with more simple structures

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69) Observe the structural formula of the organic compound given below and answer accordingly:



(1) It represents trihydroxypropane

(2) It represents cholesterol-A component of cell membrane

(3) It represents a phospholipid called gingelly oil

(4) It represents a phospholipid called lecithin, a component of cell membrane

70) Which component of lipids is responsible for the formation of esters in triglycerides?

(1) Fatty acid and phosphate

(2) Fatty acid and glycerol

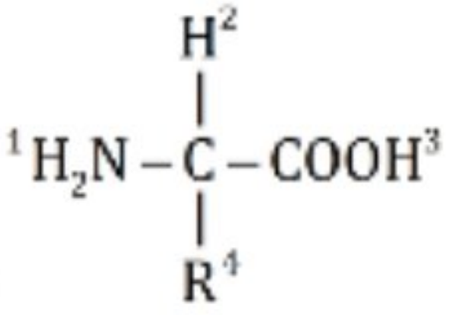
(3) Glycerol and phosphate

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(4) Phosphorus and fatty acid

71) Which of the two groups (marked as 1, 2, 3 & 4) of following formula involved in peptide bond



between different amino acids?

(1) 2 and 3

(2) 1 and 3

(3) 1 and 4

(4) 2 and 4

72) A dehydration reaction links two glucose molecules to produce maltose. If the formula for glucose is $C_6H_{12}O_6$ then what is the formula for maltose?

(1) $C_{12}H_{24}O_{12}$

(2) $C_{12}H_{22}O_{11}$

(3) $C_{12}H_{24}O_{11}$

(4) $C_{12}H_{20}O_{10}$

73) Given below are the example of few amino acids :
Glutamic acid, lysine, valine, tyrosine, phenylalanine, tryptophan.
How many of the above are aromatic amino acids?

(1) 4

(2) 2

(3) 5

(4) 3

74) **Statement I:** A protein is imagined as a line, the left end represented by first amino acid (c-Terminal) and the right end represented by the last amino acid (N-Terminal)
Statement II: Adult human haemoglobin, consists of 4 subunits (two subunits of α type and two subunits of β type)
In the light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are true

(2) Both statement I and statement II are false

(3) Statement I is true but statement II is false

(4) Statement I is false but statement II is true

75) **Assertion (A):** Peptide bond is formed between the –COOH group of one amino acid and $-NH_2$ group of another amino acid molecule.
Reason (R): A molecule of H_2O is added in this process.

(1) Assertion and Reason both are true and the reason is the correct explanation of the Assertion.

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(2) Assertion and Reason both are true but reason is not a correct explanation of the Assertion.

(3) Assertion is true but reason is false.

(4) Assertion and Reason both are false.

76) Identify the correct match from column I and II.

	Column I		**Column II**
(i)	GLUT 4	(a)	Disaccharide
(ii)	Trypsin	(b)	Hormone
(iii)	Insulin	(c)	Enzyme
(iv)	Maltose	(d)	Enables glucose transport into cells

(1) i-(d), ii-(c), iii-(b), iv-(a)

(2) i-(a), ii-(b), iii-(c), iv-(d)

(3) i-(b), ii-(a), iii-(d), iv-(c)

(4) i-(c), ii-(b), iii-(d), iv-(c)

77) Fibroin of silk exhibit _____ structure of protein

(1) Primary

(2) Secondary

(3) Tertiary

(4) Quarternary

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78) How many peptide bonds present in a 900 amino acid containing polypeptide chain?

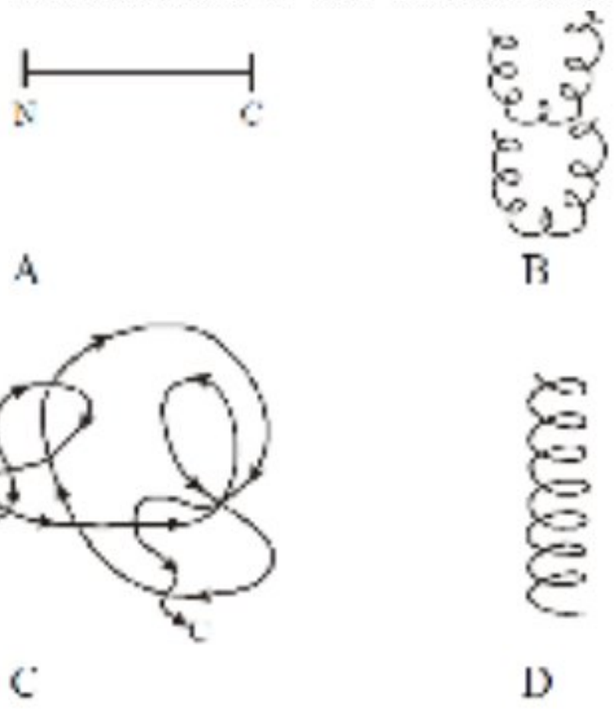
(1) 175

(2) 450

(3) 879

(4) 899

79) What kinds of the structures of proteins are shown in the given figure (A, B, C and D)?



(1) A = 1º structure, B = 2º structure, C = 3º structure, D = 4º structure

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(2) A = 4º structure, B = 2º structure, C = 3º structure, D = 1º structure

(3) A = 1º structure, B = 4º structure, C = 3º structure, D = 2º structure

(4) A = 4º structure, B = 3º structure, C = 2º structure, D = 1º structure

80) Which of the following are all nucleotides?

(1) Adenosine, Cytidylic acid, Cytosine

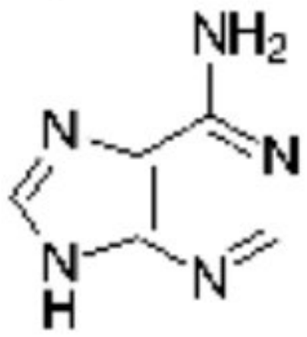
(2) Adenylic acid, Cytidylic acid, Guanylic acid

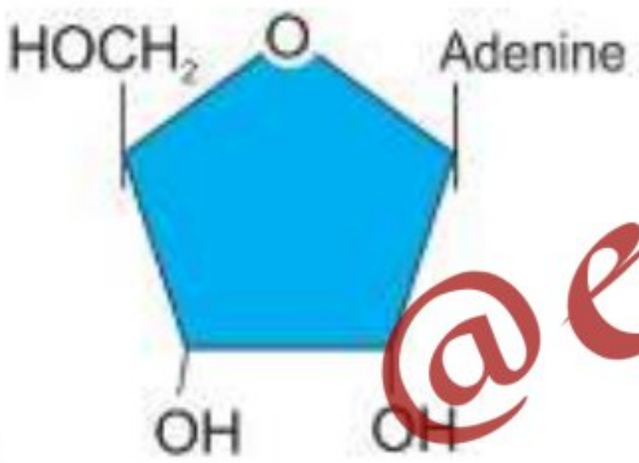
(3) Cytidine, Adenine, Adenylic acid

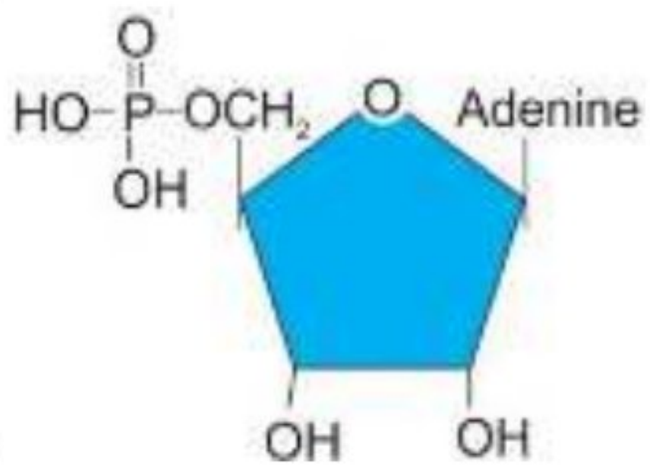
(4) Uracil, Thymidine, Thymidylic acid.

81)

Identify the structural formulae and select correct option.

(A) 

(B) 

(C) 

	A	B	C
(1)	Adenine	Adenosine	Adenylic acid
(2)	Cytosine	Adenosine	Adenylic acid
(3)	Adenosine	Adenylic acid	Adenine
(4)	Uracil	Adenosine	Adenylic acid

(1) 1

(2) 2

(3) 3

(4) 4

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82) Which of the following bond is not related to nucleic acid ?

(1) H-bond
(2) Ester-bond
(3) Glycosidic bond
(4) Peptide bond

83) The two strands of DNA are held together by :-

(1) Peptide bonds
(2) S–S bonds
(3) Hydrogen bonds
(4) Phosphodiester bonds

84) Find the correct statement about phosphodiester linkage between adjacent nucleotides in nucleic acids

(1) 3'-phosphate of one nucleotide joins the 3'-hydroxyl of the next nucleotide
(2) 3'-phosphate of one nucleotide joins the 5'-hydroxyl of the next nucleotide
(3) 5'-phosphate of one nucleotide joins the 5'-hydroxyl of the next nucleotide
(4) 3'-hydroxyl of one nucleotide joins the 5'-phosphate of the next nucleotide

85) Which of the following statements is true regarding the chemical composition of living tissues?

(1) Lipids are considered strict macromolecules.
(2) The acid insoluble pool represents the cytoplasmic composition.
(3) The acid soluble pool roughly represents the cytoplasmic composition.
(4) Water is the least abundant chemical in living organisms.

86) Which of the following amino acid's 'R' group could be a hydrogen.

(1) Glycine
(2) Valine
(3) Lysine
(4) Tyrosine

87) A nucleoside differs from a nucleotide is not having :-

(1) Phosphate
(2) Sugar
(3) Nitrogen base
(4) Amino acid

88) The four elements which make up 95% of all elements found in a living system are –

(1) C, H, O, N

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(2) C, H, O, P

(3) C, H, O, S

(4) C, N, O, P

89) The K_m value of the enzyme is the value of the substrate concentration at which the reaction reaches to -

(1) Zero

(2) 2 V_{max}

(3) $\frac{1}{2}V_{max}$

(4) $\frac{1}{4}V_{max}$

90) **Assertion (A):** The protein part of enzyme is called apoenzyme.
Reason (R): Zinc acts as prosthetic group of the enzyme carboxypeptidase.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	2	4	4	1	3	2	1	1	3	3	3	1	1	2	4	2	4	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	2	2	4	2	2	3	2	4	2	2	2	3	1	3	3	4	2	4	3
Q.	41	42	43	44	45															
A.	2	3	3	3	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	4	3	2	2	4	2	3	4	4	2	4	3	4	2	4	4	2	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	1	1	4	1	3	2	3	3	4	3	3	3	4	2	4	4	4	2
Q.	86	87	88	89	90															
A.	2	3	2	2	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	4	3	3	3	2	2	4	3	4	4	2	3	2	4	3	4	4	1	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	4	2	3	3	4	2	3	1	1	3	3	3	1	2	1	4	1	1	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	4	2	3	2	1	2	4	4	3	3	4	1	3	2	2	1	2	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	2	4	4	1	3	3	3	4	2	2	2	4	4	3	1	2	4	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	4	3	4	3	1	1	1	3	3										

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SOLUTIONS

PHYSICS

1)

Given, the velocity of projection as $\vec{v} = a\hat{i} + (b - ct)\hat{j}$.....(i)

Or, $\vec{v} = v_x\vec{i} + v_y\hat{j}$

And we know that $v_x = u_x$ and $v_y = u_y - gt$

$\Rightarrow V = u_x\hat{i} + (u_y - gt)\hat{j}$.....(ii)

Comparing (i) & (ii) we have

$u_x = a$, $u_y = b$ and $g = c$

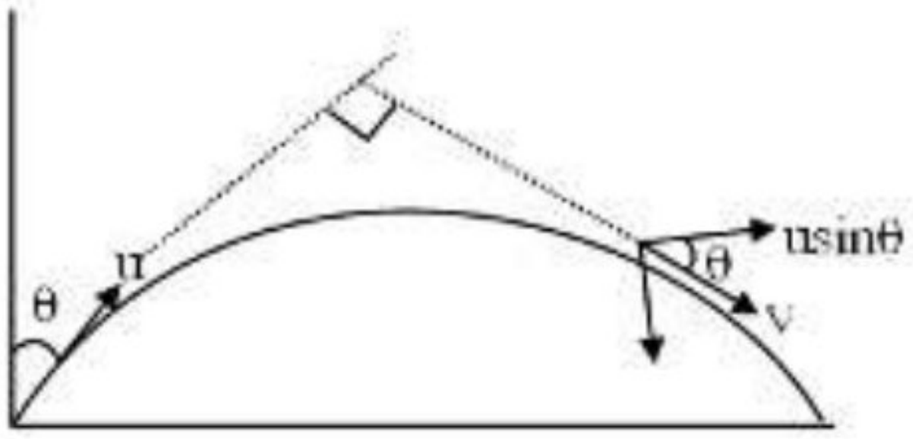
$\square$ Range is given by, $R = \frac{2u_xu_y}{g} = \frac{2ab}{c}$

2) $H = \frac{U^2\sin^2\theta}{2g}$, $H \propto \sin^2\theta$

$\frac{H_1}{H_2} = \frac{\sin^2 30^\circ}{\sin^2 60^\circ} = \frac{(1/2)^2}{(\sqrt{3}/2)^2} = \frac{1}{3}$

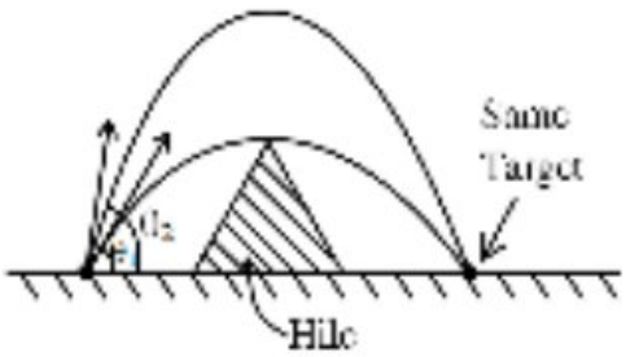
3) Sol. $T_1T_2 = \frac{2u\sin\theta}{g} \times \frac{2u\sin(90-\theta)}{g} = \frac{2R}{g}$

4) Ans. (4)



$v\cos\theta = u\sin\theta$

$(v = u\tan\theta)$



5)

For same range with same velocity of projection

$\theta_1 + \theta_2 = \pi/2$

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$$\theta_2 = \frac{\pi}{2} - \frac{5\pi}{36} = \frac{13\pi}{36}\text{rad}$$

6)

$\because R_P = R_E$

$$\frac{U_P^2 \sin 2\theta}{g_P} = \frac{U_E^2 \sin 2\theta}{g_E}, \frac{(3)^2}{g_P} = \frac{(5)^2}{9.8}$$

$$\therefore g_P = \frac{9 \times 9.8}{25} = 3.5\ m/s^2$$

7)

Question Explanation : Asking about the average velocity of the particle between its points projection & highest point of trajectory.

Concept : This question is based on average velocity in projectile motion.

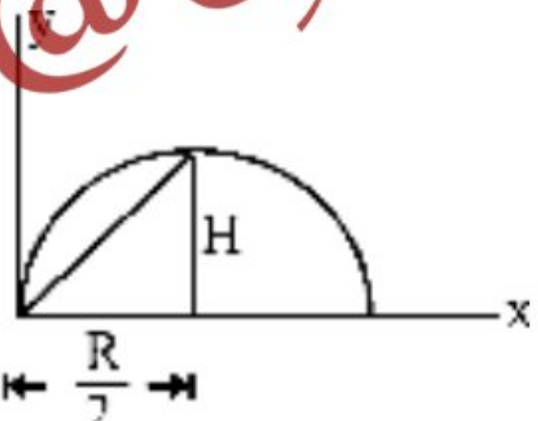
Solution :

$$\text{Average velocity} = \frac{\text{Displacement}}{\text{Time}}$$

$$v_{avg} = \frac{\sqrt{H^2 + \frac{R^2}{4}}}{T/2}$$

Here. H = maximum height

$$= \frac{v^2 \sin^2\theta}{2g}$$



$$R = \text{range} = \frac{v^2 \sin 2\theta}{g} = \frac{R}{2}$$ and T = time of flight = $\frac{2v \sin\theta}{g}$

$$\therefore \quad v_{avg} = \frac{v}{2}\sqrt{1 + 3\cos^2\theta}$$

Alternate solution :

$$\langle V_x \rangle = \frac{V\cos\theta + V\cos\theta}{2} = V\cos\theta$$

$$\langle V_y \rangle = \frac{0 + V\sin\theta}{2} = \frac{V\sin\theta}{2}$$

$$\langle V \rangle = \sqrt{V_x^2 + V_y^2} = \sqrt{V^2\cos^2\theta + \frac{V^2\sin^2\theta}{4}}$$

$$\langle V \rangle = \frac{V}{2}\sqrt{1 + 3\cos^2\theta}$$

Final answer : (4)

8)

$\because x = kt \quad \therefore t = \frac{x}{k}$

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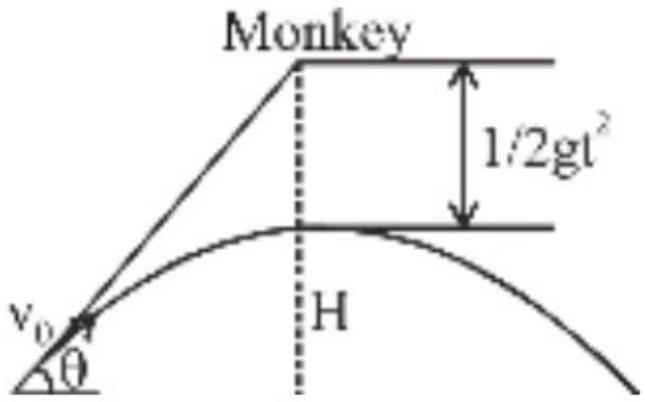
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$$\square\ y = kt - akt^2 = \quad k = \left(\frac{x}{k}\right) - \alpha k\left(\frac{x}{k}\right)^2$$

$$\square\ y = x - \frac{\alpha x^2}{k}$$

9)



$$H = v^0 \sin\theta t - \frac{1}{2}gt_2$$

The bullet will hit the monkey if the monkey falls down.

10)

Explain Question:

Find the angle of projection for a particle passing through point B and meeting a fly at C.

Concept:

Equation of trajectory and horizontal range in projectile motion.

Solution:

At point B (2, 4), trajectory is:

$$4 = 2\tan\theta - \frac{10(2)^2}{2u^2\cos^2\theta}$$

Fly speed $v_f = 2$, time t to reach C: BC = 2t

Vertical motion for projectile from B to C:

$$0 = u_y t - \frac{1}{2}gt^2$$

Horizontal distance BC for projectile:

BC = (u θ)t

By comparing horizontal speeds, u θ = 2

Substituting back into trajectory: $\tan\theta = \frac{9}{2}$

Final Answer: Option (1)

11)

u_x should be same.

12) **Explain the Question :** Two balls, A and B, are thrown from a tower of height H. Ball A is thrown with $10\sqrt{2}$ m/s at 45° upward from the horizontal, and Ball B is thrown with 20 m/s at 30° upward from the horizontal. determine which ball hits the ground first.

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Concept : This question is based on projectile motion - oblique projection.
Formula :

$$s_y = u_y t + \frac{1}{2} g t^2$$

Solution : The **time of flight (t)** for a projectile to hit the ground is governed **only by the vertical motion**. Since both balls start at the same height H and undergo the same constant vertical acceleration (g), the time of flight depends only on their initial vertical velocity component (u_y).

1. **Ball A**. $U_A = 10\sqrt{2}$ m/s at $\theta_A = 45°$ **upward**.

$$u_{yA} = U_A \sin(\theta_A) = 10\sqrt{2} \sin(45°) = 10\sqrt{2}\left(\frac{1}{\sqrt{2}}\right) = +\ 10\text{m/s}$$

2. **Ball B** : $U_B = 20$m/s at $\theta_B = 30°$ **upward**

$$u_{yB} = U_B \sin(\theta_B) = 20 \sin(30°) = 20\left(\frac{1}{2}\right) = +\ 10\text{m/s}$$

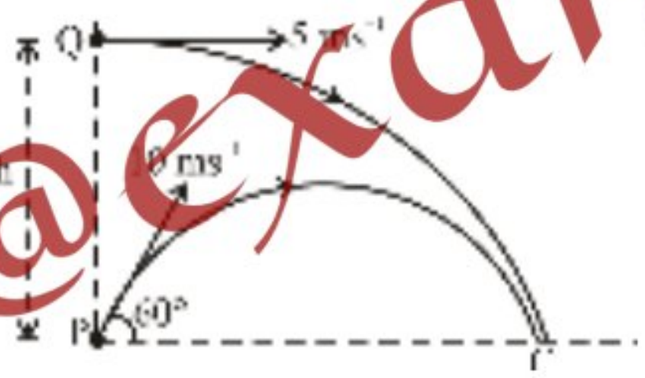
Since the initial vertical velocities are identical ($u_{yA} = u_{yB} = 10$ m/s) and they both start from the same height H, the time of flight (t) will be the same for both balls, as shown by the quadratic equation for t:

$$H = -(10)t + \frac{1}{2} g t^2$$

Therefore, both balls will strike the ground at the same time.
Final answer : 3

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13) **Explaining :**



ask : from what height h should another particle Q be projected with velocity 5 m/s so that both collide at point c.
Concept : This question is based on horizontal projection from height
Solution :
Time of fall of Q = time of flight of P

$$\sqrt{\frac{2h}{g}} = \frac{2u \sin\theta}{g}$$

Put u = 10 m/s, $\theta = 60°$, g = 10 m/s^2

$$\sqrt{\frac{2h}{10}} = \frac{2 \times 10 \sin 60°}{10}$$

$\Rightarrow$ h = 15 m
Final Answer : (3)

14) $v \sin 37º = 100 \Rightarrow v = \frac{500}{3}$ m/s

$$1500 = 100t + \frac{1}{2} \times 10 \times t^2$$

$$\Rightarrow \frac{t^2}{2} + 10t - 150 = 0$$

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$$\Rightarrow t = \frac{-20 \pm 40}{2}$$

$t = -30$ or $10s$

$$R = v\cos 37^\circ \times 10 = \frac{4000}{3}$$

15) **Question Explanation :** Find the time when initial velocities u_1 and u_2 became mutually perpendicular.

Concept: Projection from height (Horizontal Projection)

Solution:

Velocity of first particle at time t $\vec{v}_1 = u_1\hat{i} - gt\hat{j}$

Velocity of second particle at time t $\vec{v}_2 = -u_2\hat{i} - gt\hat{j}$

According to question at time t $\vec{v}_1 \perp \vec{v}_2 \Rightarrow \vec{v}_1.\vec{v}_2 = 0$

$-u_1 u_2 + g^2 t^2 = 0$

$g^2 t^2 = u_1 u_2$

$$t^2 = \frac{u_1 u_2}{g^2} \Rightarrow t = \frac{\sqrt{u_1 u_2}}{g}$$

Final Answer: Option (1).

16) **Explain Question:**

Finding the angle of projection for a projectile given its final velocity.

Concept:

Kinematics and Projectile Motion.

Solution:

The horizontal component of velocity remains constant. We use the final velocity vector to find the initial components and calculate the launch angle.

- Final Velocity: $v = 10\hat{i} - 10\hat{j} \Rightarrow v_x = 10, v_y = -10$
- Initial horizontal velocity (u_x): $u_y = v_x = 10$ m/s
- Initial vertical velocity (u_y): $v_y = u_y - gt \Rightarrow -10 = u_y - (10 \times 1.5) \Rightarrow u_y = 5$m/s
- Angle of projection (θ): $\tan\theta = \frac{u_y}{u_x} = \frac{5}{10} = \frac{1}{2}$

Final Answer: Option (2)

17)

Explain the Question : A bus accelerates at 1 m/s^2 ; a man 48 m behind runs at 10 m/s. Find time to catch bus.

Concept : This question is based on equate positions : displacement of man = displacement of bus + initial separation.

Solution : Bus displacement : $s_b = \frac{1}{2}at^2 = 0.5t^2$

Man displacement : $s_m = 10t$

Condition : $s_m = s_b + 48 \Rightarrow 10t = 0.5t^2 + 48$

$0.5t^2 - 10t + 48 = 0 \Rightarrow t^2 - 20t + 96 = 0$

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$t = \frac{20 \pm \sqrt{400-384}}{2} = \frac{40 \pm 4}{2} = 8$ or 12s

Smaller positive time = 8s.

Final answer : 4

18) $\bar{V}_{CT} = \bar{V}_C - \bar{V}_T = V\hat{j} - V\hat{i} = V\left(-\hat{i} + \hat{j}\right)$

19)

Relative velocity between either car (1st or 2nd) and 3rd car = u + 30

where

u = velocity of 3rd car

Relative Displacement = 5 km

Time interval = 4 min.

$\Rightarrow u + 30 = \frac{5}{4}$ km/min

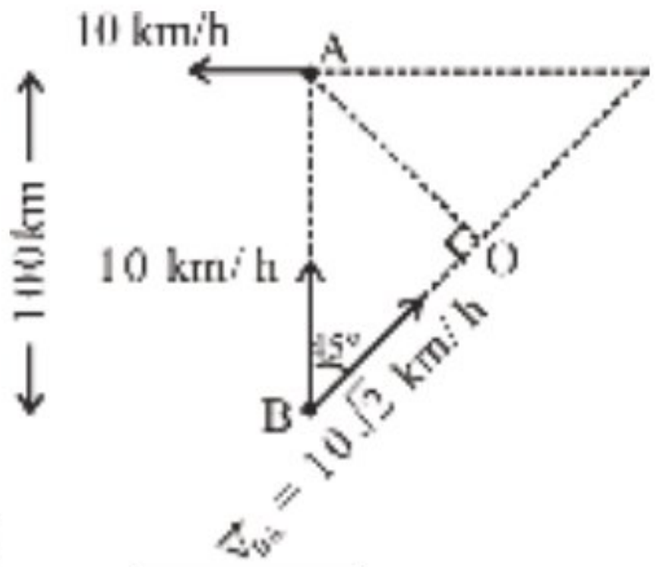
$= \frac{5 \times 60}{4}$ km/h.= 75

$\Rightarrow$ u = 45 km/h

20) speed of man $(v_m) = \frac{d}{90}$

speed of escalator $(v_s) = \frac{d}{60}$

$t = \frac{d}{v_m + v_s} = \frac{d}{\frac{d}{90} + \frac{d}{60}} = \frac{60 \times 90}{150} = 36$ sec



21)

$|\vec{v}_{BA}| = \sqrt{10^2 + 10^2} = 10\sqrt{2}$ kmph

Distance OB = 100 cos45°= $50\sqrt{2}$km

Time taken to reach the shortest distance between A & B = $\frac{50\sqrt{2}}{|\vec{v}_{BA}|} = \frac{50\sqrt{2}}{10\sqrt{2}}$

t = 5 hrs.

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22)

Explain Question : Asking about the true direction of car which was absened by two person A & B.

Concept : This question is based on relative motion in a plane.

Solution :

$\vec{v}_B = 2\vec{v}_A$

$\vec{v}_{CA} = x\hat{i} \Rightarrow \vec{v}_C - \vec{v}_A = x\hat{i}$ ... (1)

$\vec{v}_{CB} = x\hat{j} \Rightarrow \vec{v}_C - \vec{v}_B = x\hat{j}$

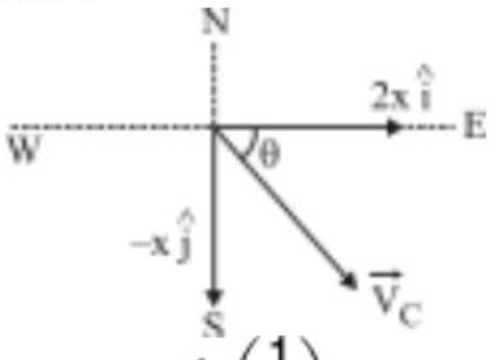
$\Rightarrow \vec{v}_C - 2\vec{v}_A = x\hat{j}$ ...(2)

Multiply eq. (1) by (2) and subtract eq. (2) from it

$2\vec{v}_C - 2\vec{v}_A - (\vec{v}_C - 2\vec{v}_A) = 2x\hat{i} - x\hat{j}$

$\vec{v}_C = 2x\hat{i} - x\hat{j}$

dir^n. $= \tan^{-1} \rightarrow \tan\theta = \frac{x}{2x} = \frac{1}{2}$



$\theta = \tan^{-1}\left(\frac{1}{2}\right)$

$\tan^{-1}\left(\frac{1}{2}\right)$ south of east

Final answer : 2

23)

To catch $(v_y)_A = (v_y)_B$

$\Rightarrow v\sin 37° = 30$

$\Rightarrow v = \frac{30}{3/5} = 50$ m/s

Time taken $= \frac{200}{v\cos 37°} = 5$ s.

24)

Question Explanation : Asking about the correct statement for the collision between the two projectile as shown in Diagram.

Concept : Collision between projectiles.

Solution :

For the collision $(u_y)_{relative}$ should be zero as the $\vec{r}_{BA} = 80\hat{i}$
So that

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For collision $u_{y_1} = u_{y_2}$

$50\sin\theta = 40\sqrt{2}\sin 45°$

$\sin\theta = \frac{4}{5}$

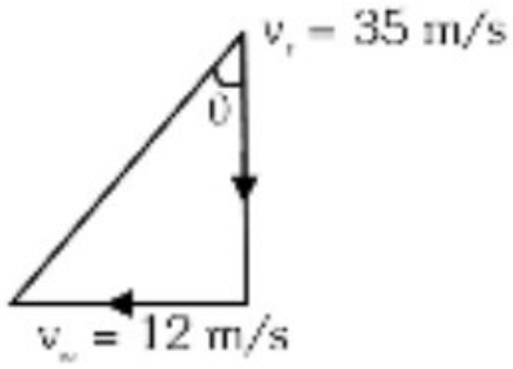
$\theta = 53°$

. **Final Answer:** option (4)

25) $\vec{V}_{rm} = \vec{V}_r - \vec{V}_m = -3\hat{j} - 3\hat{i}$

$\left|\vec{V}_{rm}\right| = 3\sqrt{3}$ m/s

$\tan\theta = 1, \theta = 45°$



26) $\tan\theta = \frac{12}{35}$

$\Rightarrow \theta = \tan^{-1}\left(\frac{12}{35}\right)$ w.r.t rain

27) The velocity upstream is (4 – 2) km/hr and downstream is (4 + 2) km/hr.

 Total time taken $= \frac{2}{2} + \frac{2}{6} = \frac{4}{3}$h

= 80 minutes

28) $x = \ell'_A - \ell'_B$ = const at any temperature

it means $\Delta l_A = \Delta l_B$

$\Rightarrow \ell_A\alpha_A\Delta T = \ell_B\alpha_B\Delta T$

$\Rightarrow \ell_A\alpha_A = \ell_B\alpha_B$

29) $\frac{T_X - (20°X)}{220°X - 20°X} = \frac{T_Y - (-40°Y)}{120°Y - (-40°Y)}$

$\frac{T_X - 20°X}{200°X} = \frac{T_Y + 40°Y}{160°Y}$

$\frac{100°X - 20°X}{200°X} = \frac{T_Y + 40°Y}{160°Y}$

$\frac{80°X}{200°X} \times 160Y = T_Y + 40°Y$

$T_Y = 24°Y$

30)

$\gamma_{apparent} = \gamma_{actual} - \gamma_{container}$

For Copper Vessel : $C = \gamma - 3A$

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For Silver Vessel : $S = \gamma - 3x$

$$x = \frac{C + 3A - S}{3}$$

31) 60% of $\frac{dQ}{dt} = ms\frac{dT}{dt}$

$$\frac{60}{100} \times 5000 = 10 \times 4200 \times \frac{(60 - 10)}{t}$$

$$t = \frac{10 \times 4200 \times 50 \times 100}{5000 \times 60} \text{ sec} = 700 \text{ sec}$$

$= 11.6$ min

32) $\Delta P.E_{loss} = \Delta Q$ (joule)

$mgh_1 - mgh_2 = ms\Delta\theta$

$$\Delta\theta = \frac{g(h_1 - h_2)}{S}$$

$$\Delta\theta = \frac{10(10 - 5.4)}{460}$$

$\Delta\theta = 0.1^{\circ}C$

33)

Heat gain by water = Heat loss by steam

$\Rightarrow MS_w (80 - 16) = m S_w (100 - 80) + mL$

$\Rightarrow 1400 \times 1 \times (64) = m \times 1 \times (20) + m \times 540$

$\Rightarrow m = 160$ gram.

34) **Explain question :** We have to calculate temperature at the junction of copper and steel.

Concept : This question is based on law of conduction (heat current in steady state)

Solution :



Let temprature of junction is T

at steady state

$$\frac{100 - T}{R_{Cu}} = \frac{T - O}{R_{Steel}}$$

$$\Rightarrow \frac{100 - T}{\frac{18}{(9K)A}} = \frac{T}{\frac{6}{(K)A}}$$

$300 - 3T = T$

$4T = 300 \Rightarrow T = 75°C$

Final answer : Option(1)

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35) $\frac{T_B-90}{R}+\frac{T_B-0}{\frac{R}{2}}+\frac{T_B-20}{\frac{R}{2}}=0$

36) $-\frac{dT}{dx}\propto\frac{1}{A}$

37) $\frac{20}{1}=(k)\left[\frac{60+80}{2}-30\right]\Rightarrow k=\frac{1}{2}$

$\frac{60-50}{t}=k\left[\frac{60+50}{2}-30\right]=\frac{1}{2}[25]$

$t=\frac{20}{25}\min=\frac{4}{5}\min=48\text{ sec.}$

38) **Explain Question:** To find radiating power

Concept: Stefan - Boltzmann law of b/ac/c body radiation, Rate of Heat loss and rate of cooling.

Solution: $\frac{60}{E}=\frac{(1000)^4-(500)^4}{(1500)^4-(500)^4}$

$\frac{60}{E}=\frac{(500)^4\left[2^4-1\right]}{(500)^4\left[3^4-1\right]}=\frac{15}{80}$

E = 320 W

Final Answer: 320 W

39) $\frac{E}{E'}=\frac{\sigma(400)^4\times 8\times 4}{\sigma(600)^4\times 4\times 2}$

$\frac{E}{E'}=\left(\frac{2}{3}\right)^4\times 4$

$E'=\frac{81}{64}E$

40) **Question Explain**

We are given stars of different colors and need to compare their surface temperatures based on their color: Red, Green, and Blue.

Concept Based

According to Wien's displacement law, the color of a star is inversely related to the wavelength it emits most strongly, which is inversely proportional to its temperature.

Formula Used

$\lambda_{max}T = b$ (Wien's Law)

Calculation

$\lambda_{max} \propto 1/T$

Shorter wavelength means higher temperature. Blue has the shortest wavelength, followed by Green, then Red.

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So:

$$T_B > T_G > T_R$$

Answer: Option 3

$$T_B > T_G > T_R$$

41)

Question Explanation : For particular wavelength λ_0 power radiated by black body is P now for different temperature if maximum wavelength is $\frac{\lambda_0}{4}$ then power radiated will be.

Concept : This question is based on wein's law.

Solution :

From stefan's law $P \propto T^4$ .....(1)

and from wein's law $\lambda_m \propto \frac{1}{T}$ .....(2)

by eq. (1) & (2)

$$P \propto \frac{1}{\lambda^4}$$

$$\frac{P_2}{P_1} = \left(\frac{\lambda_1}{\lambda_2}\right)^4$$

$\lambda_1 = \lambda_0$ & $\lambda_2 = \frac{\lambda_0}{4}$

$$\frac{P_2}{P} = \left(\frac{\lambda_0}{\frac{\lambda_0}{4}}\right)^4$$

$P_2 = (4)^4 P$

$P_2 = 256 P$

Final answer : Option (2)

42)

Explanation :

We need to determine the correct graph between In $(\theta - \theta_0)$ and time (t) according to Newton's Law of Cooling.

Concept:

Newton's Law of Cooling states that the rate of cooling of a body is proportional to the temperature difference between the body and its surroundings.

Formula:

$$\frac{d\theta}{dt} = --k(\theta - \theta_0)$$

Calculation :

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$$\int \frac{d\theta}{\theta - \theta_0} = -k \int dt$$

$$\ln(\theta - \theta_0) = -kt + C$$

Taking exponetials,

$$\theta - \theta_0 = e^{-kt + c} = e^{C} e^{-kt}$$

Thus, rewriting in logarithmic form,

$$\ln(\theta - \theta_0) = -kt + C$$

$$\ln(\theta - \theta_0) = -kt + C$$

Graph Analysis

A. Th equation is of the form y = mx + c, which represents a straight -line equation.
B. The slope (–k) is negative, meaning the graph should be a straight line with a negative slope.

Conclusion:
Hence, option (3) is correct.

43) **Explain Question :**

A rigid tank holds 35 kg of nitrogen at 6 atm pressure. Oxygen is added until the total pressure reaches 9 atm (at constant temperature). We need to find the mass of the oxygen added.

Concept:

According to Dalton's Law and the Ideal Gas Law (PV = nRT), if volume and temperature are constant, the pressure is directly proportional to the number of moles ($P \propto n$).

Solution :

A. Initial Pressure due to Nitrogen $(P_{N_2}) = 6 \text{ atm}$

B. Final Total Pressure $(P_{total}) = 9 \text{ atm}$

C. Pressure due to added Oxygen $(P_{O_2}) = 9 - 6 = 3 \text{ atm}$

D. The ratio of pressures equals the ratio of moles: $\frac{P_{O_2}}{P_{N_2}} = \frac{3}{6} = 0.5$

E. Therefore, moles of $O_2 = 0.5 \times$ moles of N_2

F. Molar mass of Nitrogen $(N_2) = 28 \text{ kg/kmol}$

G. Moles of Nitrogen $(n_{N_2}) = \frac{\text{Mass}}{\text{Molar Mass}} = \frac{35}{28} = 1.25 \text{ kmol}$

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H. Moles of Oxygen $(n_{O_2}) = 0.5 \times 1.25 = 0.625$ kmol

I. Molar mass of Oxygen $(O_2) = 32$ kg/kmol

J. Mass of Oxygen = $\text{Moles} \times \text{Molar Mass} = 0.625 \times 32 = \mathbf{20}$ kg

Final Answer: Option 3

44) **Question Explanation:** Two equal-volume vessels containing ideal gas are connected. One is heated to 600 K the other kept at 300K. We must find the final common pressure at equilibrium.

Concept: Mole conservation

Solution:

Apply mole conservation

$\mu + \mu = \mu_1 + \mu_2$

$$\frac{(1)(V)}{R \times 300} \times 2 = \frac{P}{R}\left[\frac{V}{600} + \frac{V}{300}\right]$$

On solving

$$P = \frac{4}{3} \text{ atm}$$

Final answer: option (3)

45) $P_1V_1 = P_2V_2$

$(760)\ (0.1) = P_2(0.1 + 0.09)$

$P_2 = 400$ mm of Hg = 0.4 m of Hg.

Hence change in pressure

$\Delta P = P_1 - P_2 = 760 - 400 = 360$ mm of Hg

= 0.36 metre of Hg

CHEMISTRY

46)

Explain Question: Identify species having unpaired electrons.

Concept: Magnetic nature.

Solution:

1: NO_2^+ (all paired) and CaO_2 (peroxide, paired)

2: KO_2 has O_2^- (superoxide, unpaired) but Na_2O_2 has O_2^{2-} (paired)

3: KO_2 contains superoxide ion with one unpaired electron

4: CaO_2 is peroxide with all paired electrons

Final Answer: Option (3)

47)

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Explain the Question :- Identify the compound lacking an octet around central atom.
Concept :-This question is based on Electron deficiency, octet rule
Solution :- In BCl_3, boron has only 6 valence electrons (incomplete octet), making it electron deficient. Others have complete or expanded octets.
Final answer :- (4)

48) **Explain Question:**
This question asking about hypervalent compounds exceed the octet.

Concept:
Hypervalent compounds

Solution:
PCl_5, SF_6, ClF_3 → more than 8 electrons
BF_3 → less than 8 electrons → not hypervalent

Final answer: Option (3)

49)

Correct Answer is Option (2).

50)

Explain Question
Find strongest bond formed by head-on overlap.

Concept
Extent of orbital overlap

Solution
Strength of bond depends on effectiveness of overlap.

For orbitals of same shell:

p–p overlap > s–p overlap > s–s overlap

because p orbitals are more directional and overlap more effectively.

Hence strongest bond is formed by head-on overlap of 2p and 2p orbitals.

Final Answer: Option (2)

51)

Question Explanation: The question asks to calculate the total number of sigma (σ) and pi (π) bonds present in benzonitrile (C_6H_5CN).

Concept : Sigma bond, Pi bond, Benzonitrile structure, Aromatic hydrocarbons

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Solution:

Analyzing the molecular structure of benzonitrile ($C_6H_5C \equiv N$):

A. **Sigma (σ) bonds:**

A. 6 carbon-carbon ($C-C$) bonds in the benzene ring $= 6\sigma$

B. 5 carbon-hydrogen ($C-H$) bonds on the ring $= 5\sigma$

C. 1 carbon-carbon single bond attached to nitrile group ($C_{ring}-C$) $= 1\sigma$

D. 1 carbon-nitrogen bond in nitrile group ($C \equiv N$) $= 1\sigma$

E. **Total σ bonds** $= 6 + 5 + 1 + 1 = 13\sigma$

B. **Pi (π) bonds:**

A. 3 double bonds inside the benzene ring $= 3\pi$

B. 1 triple bond in the nitrile group ($C \equiv N$) contains 2π

C. **Total π bonds** $= 3 + 2 = 5\pi$

Final answer: Option (4)

52)

Explain the Question :- Identify molecule where lone pairs ≠ bond pairs on central atom.
Concept :-This question is based on Bond pairs and lone pairs around central atom, Hybridisation.
Solution :-
$H_2O \rightarrow$ 2 BP, 2 LP (equal)
$H_2S \rightarrow$ 2 BP, 2 LP (equal)
$SCl_2 \rightarrow$ 2 BP, 2 LP (equal)
$IF_5 \rightarrow$ 5 BP, 1 LP (not equal).
Final answer :- (2)

53)

Explain Question: Identify reaction where central atom's hybridisation shifts from sp^2 to sp^3.

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Concept: Hybridisation.

Solution: In BF_3 (sp^2), boron forms a coordinate bond with F^-, forming BF_4^- (sp^3) tetrahedral structure, increasing coordination number from 3 to 4.

$$\underset{sp^2}{BF_3} + F^- \longrightarrow \underset{sp^3}{BF_4^-}$$

Final Answer: Option (3)

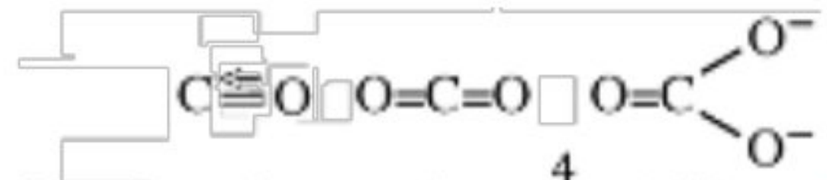
54)

fact

55) **Explain Question:** Bond length order is asked.

Concept: Bond Length.

Solution:



$$\text{Bond order} = \frac{3}{2} \quad \frac{4}{3} = 1.33$$

$$B.L. \propto \frac{1}{B.O.}$$

B. L. order $CO < CO_2 < CO_3^{-2}$

Final Answer: Option (4)

56)

Correct Answer is Option (2).

57)

Correct Answer is Option (4).

58) **Explain Question:**
π - Bond Type is asked.

Concept:
Type of π - Bond

Solution:
$SO_3 \Rightarrow sp^2$ Hyb. $\Rightarrow 1(p\pi - p\pi), 2(p\pi - d\pi)$
$SO_2 \Rightarrow sp^2$ Hyb. $\Rightarrow 1(p\pi - p\pi), 1(p\pi - d\pi)$

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ClO_4^- ⇒ sp^2 Hyb. ⇒ 3(pπ – dπ)
NO_3^- ⇒ sp^2 Hyb. ⇒ 1(pπ – pπ)

Final Answer: Option (3)

59) **Explain Question:**
The shape of a covalent molecule is determined by the arrangement of electron pairs (both bonding and lone pairs) around the central atom according to the VSEPR (Valence Shell Electron Pair Repulsion) theory. The electron pairs arrange themselves to minimize repulsion.

Concept:
VSEPR Theory

Solution:
• The molecule MX_3 has 3 bond pairs of electrons. The shape depends on the number of lone pairs on the central atom (M):

• 0 lone pairs: The electron pair geometry is trigonal planar, and the molecular shape is also triangular (or trigonal planar). Example: BF_3.

• 1 lone pair: The electron pair geometry is tetrahedral, and the molecular shape is pyramidal (specifically, trigonal pyramidal). Example: NH_3 (although this is AX_3E type). For MX_3 with one lone pair, consider a hypothetical molecule.

• 2 lone pairs: The electron pair geometry is trigonal bipyramidal, and the molecular shape is T-shape. Example: ClF_3 (AX_3E_2 type).

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Final Answer: Option (4)

60) **Explain Question:** Bond Type is asked.

Concept: Bond order.

Solution:
CaC_2 ⇒ Ca^{+2} & C_2^{-2}
Ca^{+2} & $^-C \equiv C^-$ ⇒ 1σ & 2π
Final Answer: Option (2)

61)

Correct Answer is Option (4).

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62) **Explain Question:** To determine which has maximum bond angle.

Concept: Bond Angle.

Solution:
Bond angle is determined by VSEPR theory, which minimizes electron pair repulsion around the central atom. Factors include lone pairs (increase repulsion, decrease angle), and the number of bonding pairs and their arrangement (hybridization).

• NH_4^+: Nitrogen has 4 bond pairs and 0 lone pairs (tetrahedral geometry, sp^3 hybridization). Bond angle = 109.5°.

• H_2O: Oxygen has 2 bond pairs and 2 lone pairs (bent geometry, sp^3 hybridization). Lone pair-lone pair and lone pair-bond pair repulsions reduce the bond angle to approximately 104.5°.

• H_3O^+: Oxygen has 3 bond pairs and 1 lone pair (trigonal pyramidal geometry, sp^3 hybridization). Lone pair-bond pair repulsions reduce the bond angle to approximately 107°.

• CO_2: Carbon has 2 bond pairs and 0 lone pairs (linear geometry, sp hybridization). Bond angle = 180°. (maximum bond angle)

Final Answer: Option (4)

63) **Explain Question:**
A bond angle of 90° is typically found in molecules with octahedral or square planar geometry where ligands are positioned at the corners of these shapes around a central atom.

Concept:
Bond Angle

Solution:

1. IF7: Iodine heptafluoride has a pentagonal bipyramidal geometry. Its bond angles are 90°, 72°, and angles close to 180°.

2. $SiCl_4$: Silicon tetrachloride has a tetrahedral geometry. All bond angles are approximately 109.5°.

3. XeF_4: Xenon tetrafluoride has a square planar geometry. It contains bond angles of 90° and 180°.

4. PCl_5: Phosphorus pentachloride has a trigonal bipyramidal geometry. Its bond angles are 90°, 120°, and 180°.
Therefore, $SiCl_4$ does not contain a bond angle of 90°.Answer option 2, ($SiCl_4$).

Final Answer: Option (2)

64)

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Question Explanation
We are asked which orbitals contain the unpaired electrons responsible for O_2's paramagnetism.

Concept
this question is based on Molecular Orbital Theory.

Solution

A. O_2 has 16 electrons.

B. MO filling order: σ1s2, σ*1s2, σ2s2, σ*2s2, σ2pz2, π2px2, π2py2, π*2px1, π*2py1

C. two unpaired electrons remain in π*2px , π*2py

D. These unpaired electrons cause paramagnetism.

Final Answer: Option 4

65)

Explain Question: Identify diamagnetic species containing only π bonds in vapour phase.
Concept: Molecular orbital
Solution:
C_2 configuration: $(\sigma 1s)^2(\sigma 1s)^2(\sigma 2s)^2(\sigma 2s)^2(\pi 2p_x)^2(\pi 2p_y)^2$ → bond order = 2 from π bonds only; all electrons paired → diamagnetic.
B_2 configuration: $(\sigma 1s)^2(\sigma 1s)^2(\sigma 2s)^2(\sigma 2s)^2(\pi 2p_x)^1(\pi 2p_y)^1$ → bond arises only from π orbitals in vapour phase.
Final Answer: Option (4)

66)

Both (A) and (R) are correct and (R) is the correct explanation of (A)

67)

Correct Answer is Option (3).

68)

Explain Question: Arrange HF, H_2O, and NH_3 in order of hydrogen bond strength.
Concept: This question is based on Hydrogen bonding strength ∝ electronegativity & number of H-bonds per molecule
Solution: HF forms the **strongest** individual H-bond (highest electronegativity of F), followed by H_2O, then NH_3 (least electronegative N).
Final Answer: Option (1)

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69)

Correct Answer is Option (1).

70)

Correct Answer is Option (4).

71)

Correct Answer is Option (1).

72)

Question Explanation: A concise verification of the correct solubility orders.

Concept Used: Hydration energy vs. lattice energy, and hydrogen bonding.

Answer Explanation:

A. **(i) $CaSO_4 < MgSO_4$ (True):** Smaller Mg^{2+} has a much higher hydration energy, making its sulfate more soluble than $CaSO_4$.

B. **(ii) $KHCO_3 > NaHCO_3$ (True):** $NaHCO_3$ forms a highly stable, hydrogen-bonded crystal lattice that is harder to dissolve than $KHCO_3$.

C. **(iii) $NaNO_3 < KNO_3$ (False):** Solubility decreases down the group for alkali metal nitrates; $NaNO_3$ is more soluble.

D. **(iv) LiF > LiI (False):** LiF has a massive lattice energy due to two small ions, making it less soluble than LiI.

E. **(v) $CH_3OH < CH_3OCH_3$ (False):** Methanol (CH_3OH) forms extensive hydrogen bonds with water, making it far more soluble than dimethyl ether.

Final Answer: Option (3)

73)

Explanation

A. Lattice energy is directly proportional to the product of the charges of the ions and inversely proportional to the sum of their ionic radii (Born-Landé equation). Higher charges and smaller ionic sizes lead to higher lattice energy.

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Concept

A. **Al_2O_3 > Na_2O:** Al^{3+} has a higher charge than Na^+, and O^{2-} is the common anion. Higher charge leads to higher lattice energy. **Correct.**
B. **MgF_2 < CaF_2:** Mg^{2+} is smaller than Ca^{2+}, and F^- is the common anion. Smaller ionic size leads to higher lattice energy. Therefore, MgF_2 should have higher lattice energy than CaF_2. **Incorrect.**
C. **LiF > NaF:** Li^+ is smaller than Na^+, and F^- is the common anion. Smaller ionic size leads to higher lattice energy. **Correct.**
D. **LiF > LiI:** F^- is smaller than I^-, and Li^+ is the common cation. Smaller ionic size leads to higher lattice energy. **Correct.**

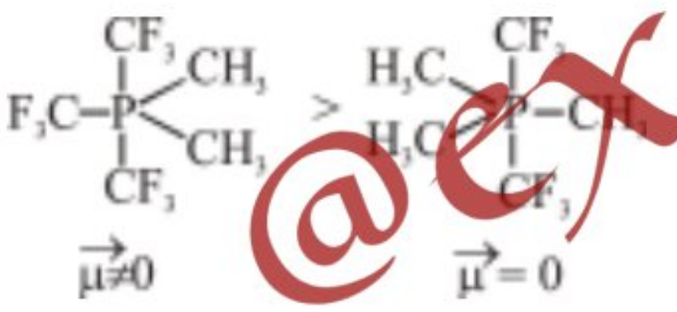
Answer option 2, (**MgF_2 < CaF_2**).

74)

Correct Answer is Option (3).

75)

Correct Answer is Option (3).

76) 

77)

Correct Answer is Option (3).

78)

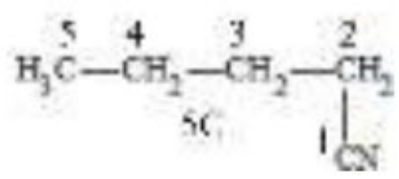
Correct Answer is Option (3).

79)

Correct Answer is Option (3).

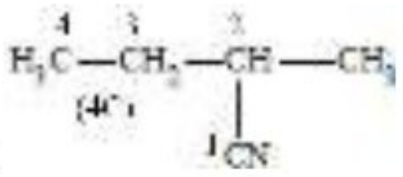
80)

Correct Answer is Option (4).

81) 

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Difference in PCC.

82)

Correct Answer is Option (4).

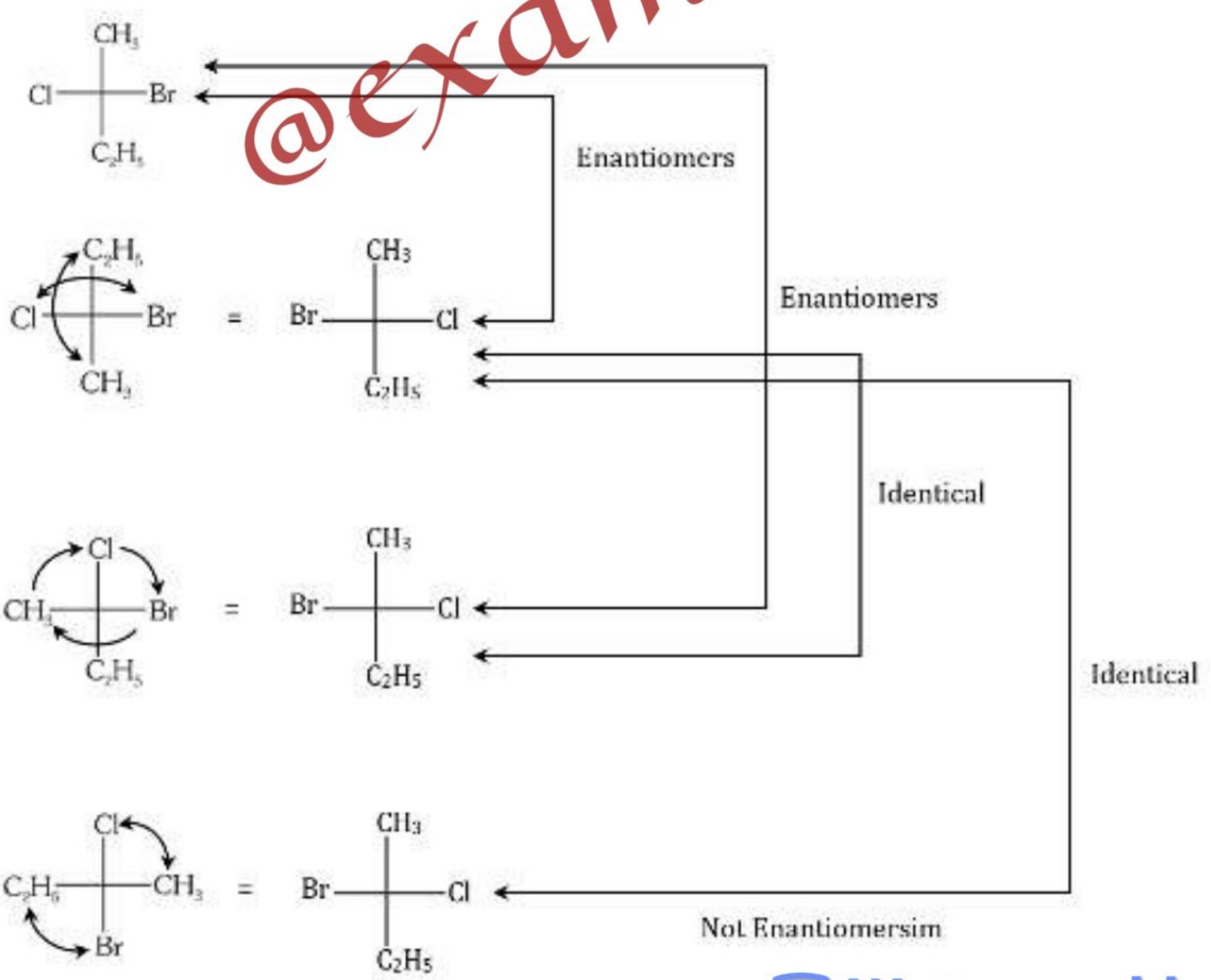
83)

Correct Answer is Option (4).

84) **Question Explanation:** Determine which stereochemical relationship statement is incorrect among four chiral molecules.

Concept: Enantiomers, Identical Structures, Configuration Comparison
Solution: Main Rule:

A. **Enantiomers:** Non-superimposable mirror images
B. **Identical:** Same configuration and spatial arrangement



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A. 1. B and C are identical □
B. 2. A and B are enantiomers □
C. 3. A and C are enantiomers □
D. 4. B and D are enantiomers □

Final Answer: Option (4)

85)

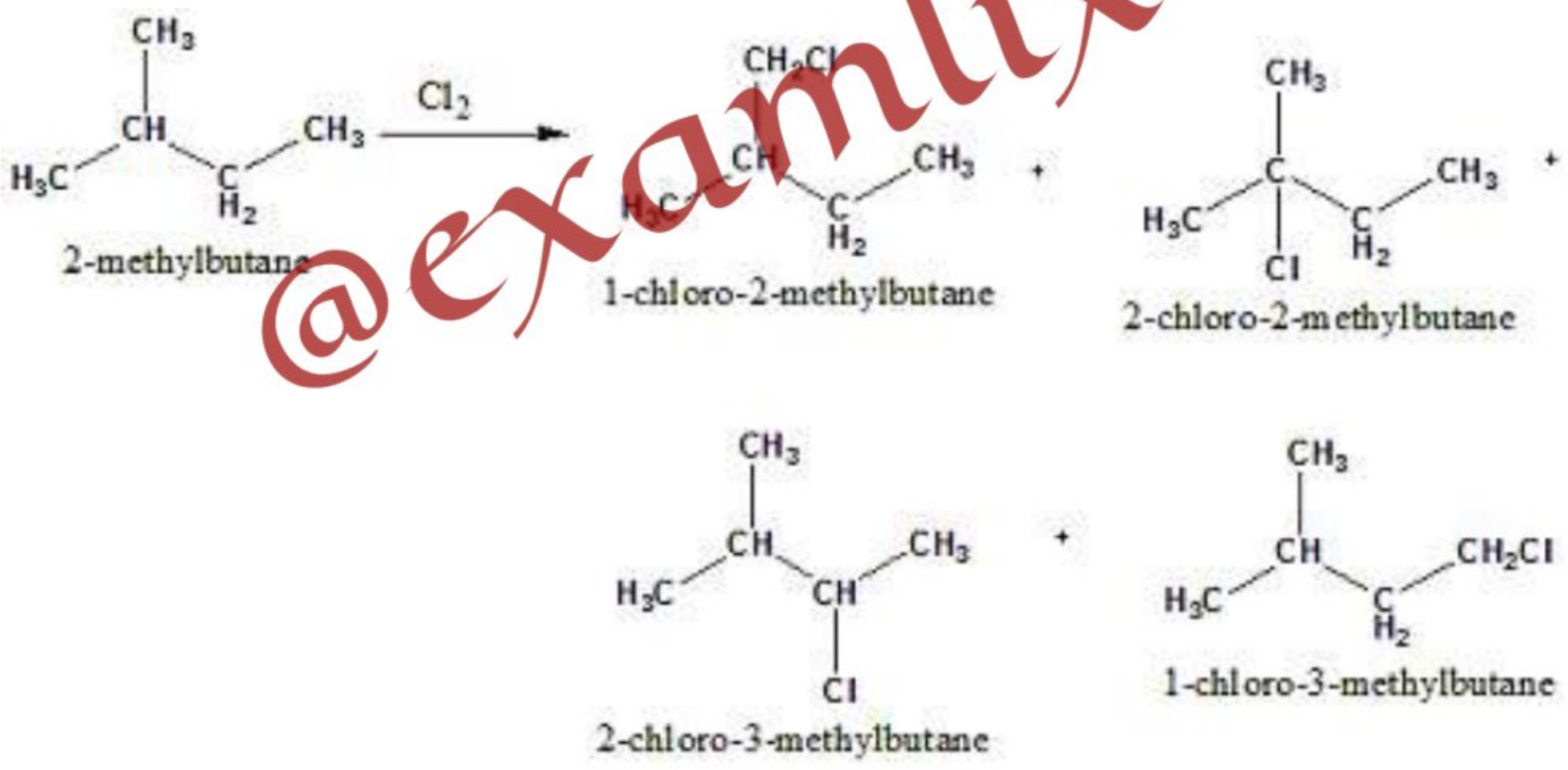
Correct Answer is Option (2).

86)

Question Explanation - Determine the number of distinct structural isomers formed by monochlorination of the given alkane structure.

Concept - Monochlorination and Structural Isomerism.

Solution -



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Final Answer - Option (2)

87)

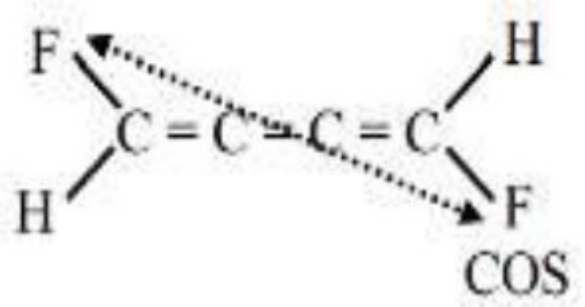
Question Explanation: Identify the compound having POS but not COS.

Concept: symmetry element POS and COS.

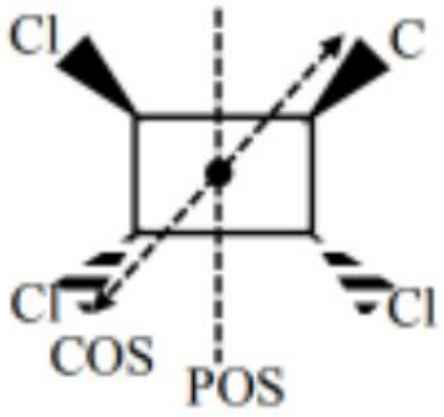
Solution:

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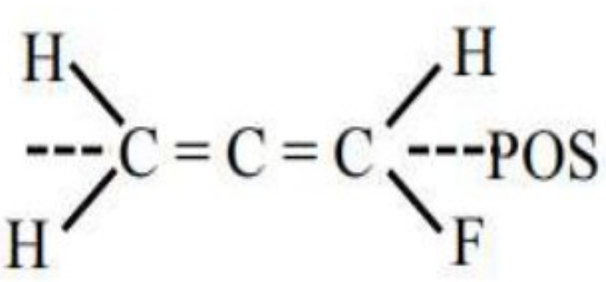
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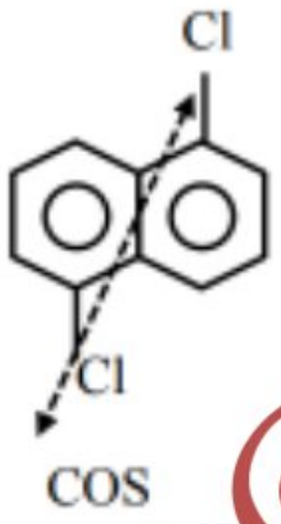
1. : Molecule is planar (1) slice cut (POS) (2 COS)



2. : Molecule is planar (1) slice cut (POS) (2 COS)



3. : Molecule is non-planar since terminal hydrogen atom are perpendicular. Hence POS is present but COS is absent.

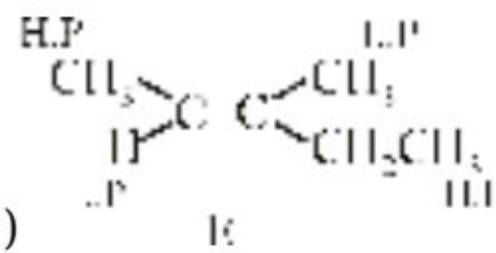


4. : Molecule is planar (1) slice cut (POS) (2 COS)

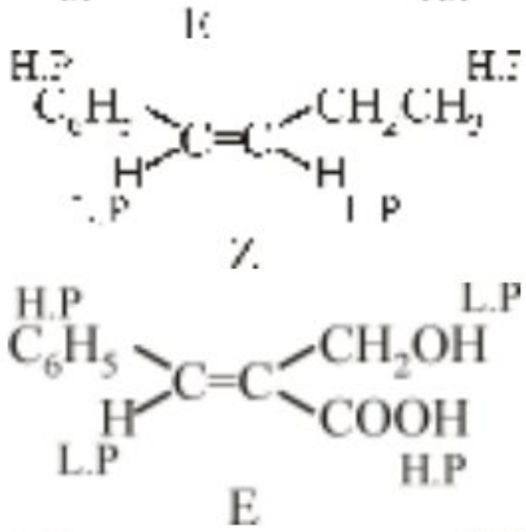
Final Answer: Option (3)

88)

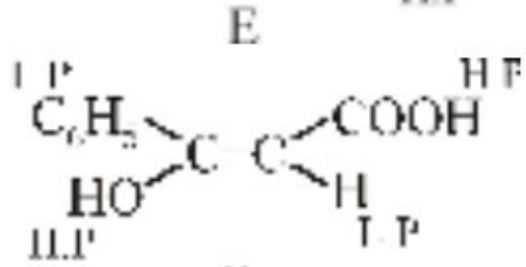
(1)



(2)



(3)



(4)

Here : H.P. → High priority group.
L.P. → Low priority group.

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89)

Correct Answer is Option (2).

Bond angle is 109°28'

90)

BIOLOGY

91)

Correct Answer is Option (4).

92)

93)

Correct Answer is Option (3).

94)

Correct Answer is Option (3).

95)

Correct Answer is Option (3).

96)

Correct Answer is Option (2).

97) Statement I is correct but statement II is false

98)

Correct Answer is Option (4).

99)

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Correct Answer is Option (3).

100)

Correct Answer is Option (4).

101)

Solution:

The correct answer is:

4. It was a phylogenetic classification

Bentham and Hooker's system is natural, based on observable features (especially flowers), not phylogenetic (not based on evolution)

102) **Chlorella and Chlamydomonas:** These are both unicellular green algae. Algae are classified under the kingdom Protista.
Paramecium and Amoeba: These are both protozoans, which are also unicellular eukaryotes. Protozoans are also classified under the kingdom Protista.

103)

Question Explanation :
Which is characteristic of viroid's and not virus.

Concept : Viroid's are RNA molecule with out protein coat.

Solution :
The correct answer is (3)
RNA molecules without protein coal.

Final Answer : option (3)

104)

Correct Answer is Option (2).

105)

Correct Answer is Option (4).

106) **Question Explanation**: Identify the organism with silica-embedded, indestructible cell walls.
Concept: Cell wall composition and durability.
Solution: The cell walls of **Diatoms** are unique. They are composed of **silica** (SiO_2), which is an extremely hard and chemically inert substance, making the walls **indestructible** and

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resistant to decomposition.
Final Answer: **Diatoms**

107)

Explain Question : The question is asking which statements (A–E) are correct about different protists.

Concept : This question is based on Protista.

Solution :
(A) **Diatoms** → chief producers of ocean -T
(B) **Dinoflagellates** → have two flagella, not without -F
(C) **Slime moulds** → saprophytic, decomposers -T
(D) **Euglenoids** → can be mixotrophic; in absence of light may ingest food like predator -T
(E) **Protozoa** → unicellular eukaryotes, not prokaryotes- F

Correct = **A, C, D.**

Final answer : option (4)

108)

Ncert page 16.

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109) NCERT-XI_Pg. # 20, 21

110)

Correct Answer is Option (4).

111)

Correct Answer is Option (4).

112)

Correct Answer is Option (4).

113)

Correct Answer is Option (2).

114)

Correct Answer is Option (3).

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NOVA

115)

Explanation :- to due to to find the depth 'd'

Concept :- This question is based on 'g' at depth 'd'

Solution :- given $g_d = \frac{g_s}{n}$

$$g_s\left[1-\frac{d}{R}\right]=\frac{g_s}{n}$$

$$1-\frac{g}{R}=\frac{1}{n}$$

$$\frac{d}{R}=1-\frac{1}{n}$$

$$d=R\left[\frac{n-1}{n}\right]$$

Answer :- 2

116)

Viruses, viroids, and lichens are not included in the five-kingdom system of classification.

The correct answer is:OPTION(4)All a, b & c

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117) NCERT (XI) Pg # 26,27

118)

Correct Answer is Option (3).

119)

Question Explanation: The question asks about identifying which of the given groups are heterotrophic organisms.

Concept: Ascomycetes

Solution: Only Ascomycetes show heterotrophic, absorptive mode of nutrition.

All others are autotrophic.

Final answer: 1 Ascomycetes

120)

Correct Answer is Option (1).

121)

Correct Answer is Option (3).

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122)

Correct Answer is Option (3).

123) NCERT (XI) Pg. # 20,26,27

124)

Correct Answer is Option (1).

125) **Explaining the Question:**
The question presents an Assertion and Reason.

Concept:
Reproduction in bacteria.

Solution:
Assertion - True
Bacteria primarily reproduce through binary fission.

Reason - True
Certain bacteria from spores specially endospores to survive adverse environment

The Reason not correctly explain Assertion. Because both statement independent.

Correct Answer:
Option (2)

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126)

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127)

Correct Answer is Option (4).

128)

Correct Answer is Option (1).

129) NCERT XI Pg # 19

130)

Correct Answer is Option (2).

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131)

Correct Answer is Option (3).

132)

NCERT XI Pg. # 128,129

133)

Correct Answer is Option (2).

134)

Question Explanation: Counting terms not related to Protista.
Concept: Characteristics of Kingdom Protista.
Solution: Terms **related to Protista**: **Cilia**, **Flagella** (locomotion), **Zygote** (sexual reproduction), **Diatomite** (diatom remains), **Photosynthesis**, **Pellicle** (Euglenoids), **Bioluminescence** (Dinoflagellates), **Decomposer** (Slime Moulds), **Holozoic** (Amoeba). (9 terms) Terms **NOT related to Protista**:

A. **Peptidoglycan wall**: Cell wall of **Bacteria** (Monera).
B. **Multicellular**: Protists are primarily **unicellular** eukaryotes.
C. **Heterocyst**: Specialized cell in some **Cyanobacteria** (Monera). (3 terms are not related to Kingdom Protista).

Final Answer: 3

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135)

NCERT-XI-Page No. # 20, 21

136)

Correct Answer is Option (1).

137)

Correct Answer is Option (2).

138)

NCERT XI, Pg. # 117

139)

NCERT Page no 158 & 159

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140)

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141)

Correct Answer is Option (3).

142) **Question Explanation**: Identifying variables that affect enzyme rate.
Concept: Factors Affecting enzyme Activity
Solution:

A. **Enzymatic activity** is highly dependent on environmental conditions.
B. **Temperature** and **pH** affect the enzyme's three-dimensional structure and active site function.
C. **Substrate concentration** determines how often the enzyme and substrate collide, affecting the reaction rate up to the point of saturation (Vmax).

Therefore, all listed factors influence enzyme activity.

Final Answer: Option 4

143) **Explain the question :** The question asks: Which of the given statements about enzymes and proteins is not correct ?
Concept : This question is based on characteristics of enzymes.
Solution : Almost all proteins are enzymes → Incorrect. Only some proteins act as enzymes; many proteins have structural, transport, or hormonal roles.
Final Answer : (1)

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144)

Question Explanation: Identify the term representing the substrate concentration at which the reaction rate becomes half of maximum velocity.

Concept: Enzyme

Solution: The correct answer is option 3 – Km (Michaelis-Menten constant).

A. Km (Michaelis-Menten constant) is defined as the substrate concentration at which the velocity of an enzyme-catalyzed reaction reaches half of its maximum velocity (Vmax). It indicates the affinity of the enzyme for its substrate (lower Km means higher affinity).

Final Answer: Option 3

145)

Correct Answer is Option (2).

146)

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Allen module

147) Allen module

148) **Solution :** -
Final Answer : (2)

149) **Solution :** a-i, b-ii, c-iii, d-iv, e-v
(a) Ions are present in relatively small amounts but are crucial for many cellular processes. They typically make up around (i) 1% of the total cellular mass.
(b) Lipids are essential for cell membranes and energy storage. They generally constitute about (ii) 2% of the cellular mass.
(c) Carbohydrates play roles in energy storage and structural support. Their percentage is usually around (iii) 3%.
(d) Nucleic acids (DNA and RNA) carry genetic information and are involved in protein synthesis. They typically make up (v) 10-15% of the cellular mass, with proteins requiring a significant amount of the cellular machinery.
(e) Proteins are the workhorses of the cell, involved in catalysis, structure, transport, and many other functions. They are the most abundant macromolecule, making up (iv) 5-7% of the cellular mass.
Final Answer : (1)

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150) Abrin

151)

Correct Answer is Option (2).

152)

Explain Question : This question is asking about the type of glycosidic bond present at the branching points of starch and glycogen.

Concept : Biomolecules – Polysaccharides and types of glycosidic linkages.

Explanation : Branching in both starch (amylopectin) and glycogen occurs through α–1,6 glycosidic linkages. The main chain has α–1,4 linkages, while branching specifically uses α–1,6 bonds.

Final answer : Option 2

153)

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154) NCERT Page no. 148

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155)

Question Explanation : Check correctness of assertion and reason regarding reducing nature of monosaccharides.

Concept : The question is based on properties of carbohydrates

Solution : The correct answer is option 1.
• Assertion is true because all monosaccharides act as reducing sugars.
• Reason is true because monosaccharides have free aldehyde or ketone group (or can form it), which enables reduction.
• Reason correctly explains assertion because presence of free functional group allows them to act as reducing agents.

Final answer : option 1

156)

Explain Question : This question is asking about the number of reducing ends in a glycogen molecule.

Concept : Carbohydrates – Structure of polysaccharides (glycogen).

Explanation : Glycogen is a highly branched polymer of glucose with only one reducing end; all other ends are non-reducing due to branching.

Final answer : Option 3

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157)

Allen module.

158) [NCERT Pg. 144]

159) NCERT pg-145

160) **Solution :** Ester bonds are formed between fatty acids and glycerol
Final Answer : (2)

161)

Correct Answer is Option (2).

162)

Explaining the Question :- The question is asking about the formula for maltose
Concept:- Carbohydrates
Solution :- In a dehydration reaction, one water molecule (H_2O)is removed; this gives the

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molecular formula of maltose ($C_{12}H_{22}O_{11}$).
$2 \times C_6H_{12}O_6 = C_{12}H_{24}O_{12} - H_2O = (C_{12}H_{22}O_{11})$
Final Answer :- Option_2

163)

Correct Answer is Option (4).

164)

Correct Answer is Option (4).

165)

Correct Answer is Option (3).

166)

Correct Answer is Option (1).

167)

Correct Answer is Option (2).

168)

Concept

A. A peptide bond forms between two amino acids.

B. The number of peptide bonds = (number of amino acids – 1).

Explanation :
Number of peptide bonds = 900 – 1 = 899.

Final Answer option 4: 899

169) **Concept :**
Solution : The type of protein structure shown in the given figures A, B, C and D are respectively primary structure, quaternary structure, tertiary structure and secondary structure.
Final Answer : (3)

170)

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Correct Answer is Option (2).

171)

Adenine: Adenine is a purine base. It has a double-ring structure.

A. It is a component of DNA and RNA.

Adenosine: Adenosine is a nucleoside.

A. It consists of adenine linked to a ribose sugar.

Adenylic acid (or Adenosine monophosphate, AMP):

Adenylic acid is a nucleotide.

A. It consists of adenine, ribose sugar, and a phosphate group.

B. (A) is Adenine, (B) is Adenosine, and (C) is Adenylic acid, **then option (1) is correct.**

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172)

NCERT Page # 106

173)

The correct answer is **3: Hydrogen bonds.**

Hydrogen bonds between the nitrogenous bases (A-T and G-C) hold the two DNA strands together.

174) **Solution/Explanation:**

3'-hydroxyl of one nucleotide joins the 5'-phosphate of the next nucleotide: This is correct. The 3' hydroxyl group of one nucleotide reacts with the 5' phosphate group of the next nucleotide, forming a phosphodiester bond.

The correct answer is option 4.

175) The text indicates that the acid soluble pool approximates the composition of the cytoplasm, while the acid insoluble fraction includes macromolecules from the cytoplasm and organelles. Lipids, although not strictly macromolecules, are part of the macromolecular fraction. Additionally, water is mentioned as the most abundant chemical in living organisms,

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not the least.

176) Ncert Page 106.

177)

The correct answer is **1: Phosphate**.

A. **Nucleoside:** Consists of a nitrogenous base linked to a sugar.
B. **Nucleotide:** Consists of a nitrogenous base, a sugar, and a phosphate group.

178)

Correct Answer is Option (1).

179)

Correct Answer is Option (3).

180)

Correct Answer is Option (3).

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